



Recent innovations in 2D magnetic materials and their potential applications in the modern era

RESEARCH: Review

Ehsan Elahi^a, Muhammad Asghar Khan^a, Muhammad Suleman^b, A. Dahshan^c, Shania Rehman^d, H.M. Waseem Khalil^e, Malik Abdul Rehman^f, Ahmed M Hassan^g, Ganesh Koyyada^{h,*}, Jae Hong Kim^{h,*}, Muhammad Farooq Khan^{i,*}

^a Department of Physics & Astronomy, Sejong University, Seoul 05006, Republic of Korea

^b Department of Nanotechnology and Advanced Materials Engineering, Sejong University, Seoul 05006, Republic of Korea

^c Department of Physics, Faculty of Science, King Khalid University, P.O. Box 9004, Abha, Saudi Arabia

^d Department of Semiconductor System Engineering, Sejong University, Seoul 05006, Republic of Korea

^e Department of Electrical Engineering, College of Engineering and Technology, University of Sargodha, Sargodha 40100, Pakistan

^f Department of Chemical Engineering, New Uzbekistan University, Tashkent 100007, Uzbekistan

^g Faculty of Engineering and Technology, Future University in Egypt, New Cairo 11835, Egypt

^h School of Chemical Engineering, Yeungnam University, Daehak-ro 280, Gyeongsan, Gyeongbuk 38541, Republic of Korea

ⁱ Department of Electrical Engineering, Sejong University, Seoul 05006, Republic of Korea

In recent years, enormous efforts have been made to identify and manipulate the exotic electrical and magnetic properties at a two-dimensional (2D) limit of various exciting materials. The spin–orbit coupling (SOC) in 2D van der Waals (vdW) heterostructures opens a fascinating and versatile platform to implement the intriguing quantum-engineered spintronic devices for practical applications. This review comprehensively outlines the modern-era progress of investigating the inherent magnetism of atomically thin 2D materials. Firstly, in this review, the most recent developments of synthesis, characterizations, functionalities, and spin textures in 2D magnetic materials are summarized in detail. Secondly, we conferred the well-known phenomena related to the proximity effect, spin–orbit torque (SOT), spin valve magnetic tunnel junction (MTJ), spin field effect transistor (FET) and magneto memristive-based applications. In addition, we also explored the possible interplay between 2D magnetic materials and associated band topology for spin caloritronics. Finally, we provided our perspective on the recent and upcoming challenges and goals of this promising research area, which leads the 2D magnetic materials to the modern era of energy-efficient quantum computing.

Keywords: Proximity effect; Magnetoresistance; Spin valve; Spin Field Effect Transistor (S-FET); Spin-Orbit Torque (SOT)

Introduction

In recent times, 2D materials have attracted much attention due to their tremendous electrical [1–3], optical [4–6], thermal [7], sensing [8] and magnetic properties [9,10] as well as their broad range of applications in the field of spintronics [11,12]. The

range of properties covered by 2D materials is now quite extensive: metallic, insulating, semiconducting, semi-metal, ferromagnetic and topological insulator [13–19]. The atoms in monolayer 2D materials are connected by covalent bonds, while the layered structures are connected by weak van der Waals (vdWs) interaction. As a result, the characteristics and adaptability of 2D-magnetic materials [20–25] have offered a distinct stand to discover new spintronic concepts with exceptional controllability. Since, this field of study based on 2D magnetic materials has

* Corresponding authors.

E-mail addresses: Koyyada, G. (ganeshkoyyada@gmail.com), Kim, J.H. (jaehkim@ynu.ac.kr), Khan, M.F. (mfk@sejong.ac.kr).

many intriguing and newly discovered properties and major potential for applications in topological magnonics [26,27], low-power spintronics [28,29], quantum computing [30,31], magneto-optics [32,33], and optical communications [27,34]. Furthermore, exotic physical phenomena at the atomically thin limit, such as the quantum anomalous Hall effect [35,36], magneto-electric multiferroics [37], and magnon valleytronics [38,39], have been discovered based on 2D magnetic materials and their heterostructure configurations. Such characteristics make 2D magnets exciting candidates for a broad scope of condensed matter physics especially in spintronic applications. Layered material exhibits intrinsic ferromagnetism, which offers an intriguing setting for research into devices that can localized spin and basic 2D magnetism [21,40,41]. Van-der Waals (vdW) magnetic materials have shown great promise for many useful applications due to their charming properties such as carrier mobility, mechanical strength and thermal conductivity [9,42–49], since the first experimental development of intrinsic ferromagnetism in monolayer CrI₃ [25] and bilayer Cr₂Ge₂Te₆ [50] vdW magnetic materials. The field of 2D magnetic materials is expanding fast, and numerous forms of 2D ferromagnetic (FM) magnets have been found and produced such as CrBr₃ [51], Cr₂Ge₂Te₆ [50], Fe₅-GeTe₂ [52], Fe₃GeTe₂ [53,54], monolayer VSe₂ [21], and antiferromagnetic (AFM) magnets such as CrCl₃, MnBi₂Te₄ [55], CrPS₄ [56], MPX₃ (M = Mn, Fe, Ni; X = S, Se) [20,57,58] etc. Specifically, the 2D magnetic insulator CrI₃ [59] presents an intriguing layer thickness-dependent magnetic order [25]: single-layer CrI₃ shows ferromagnetic behavior, however, bilayer CrI₃ represents antiferromagnetic performance. Upon the application of magnetic field of 1 Tesla, bilayer CrI₃ can be altered to ferromagnetic via spin-flip transition. Due to weak interlayer interaction, recent reports have shown antiferromagnet to ferromagnetic switching in bilayer CrI₃ by either electrostatic doping [60,61] or pure electric field [62] under a steady magnetic field.

These materials with unique spin-dependent characteristics offer a potential platform for the investigation of novel quantum phenomena and the development of innovative spintronic devices. Nowadays, several manufacturing processes are now available, including numerous renowned and cost-effective ones such as mechanical exfoliation [63,64], liquid exfoliation [65–67], chemical vapor deposition [68,69], and molecular beam epitaxy (MBE) [70,71]. The diverse phenomenon of the proximity effect has revolutionized the field of spintronics and 2D magnetic materials. It can acquire new features like superconductivity [72–74], magnetic properties [75], topological non-triviality [76], or better spin-orbit coupling [76,77]. These proximity effects not only have potential to overcome many limitations of the conventional doping or functionalization techniques of material design, but they may also be used in addition to them. In order to control superconducting [78,79], spintronics [80,81], excitonic [82,83] and topological phenomena [84,85] in heterostructures, magnetic proximity effects are essential. These effects are extremely dependent on the interfacial electrical characteristics, such as band alignment and electron wavefunction overlap. There are now more opportunities to study proximity effects in vdW heterostructures because of recent expansion of magnetic 2D materials [25,42,86]. Particularly, multilayer antiferromagnetism in which neighboring ferromagnetic monolay-

ers are antiferromagnetically related is shown in atomically thin CrI₃ [25]. Novel Applications including spin field effect transistor (S-FET), spin valve and SOT have played a vital role in making enhancements in spintronic-based industry. Spin FETs are first offered by Das and Datta in 1990 [87]. The spin-based devices imply non-volatile storage, energy efficient and quicker operation than conventional transistors [88–90]. To fabricate spin transistors, several techniques have been adopted, especially vdW heterostructures [91,92]. All these devices are reliant on spin injection and detection at ferromagnetic electrodes and voltage control of spin current in a semiconductor medium. But such devices confront substantial problems in attaining a greater high-low conductance ratio due to ineffective spin injection and relaxation in the channel materials [93–95]. Spin-orbit torque (SOT), a novel structure for quick and energy-efficient magnetic memory systems, has attracted interest as a method for current-induced regulation of magnetization in FM materials. VdW materials and their heterostructures have undergone fast recent development, opening new possibilities for better spintronic capabilities. The vdW materials with a smooth surface were found to contain a sizable set of topological materials. Additionally, in recent times, those FM vdW materials have been identified [25,96] where they showed perpendicular magnetic anisotropy (PMA) [25] and a reasonably high Curie temperature (T_C), both of which are crucial for FM layers in spintronic devices. As a result, electric multiferroics, and magnon valleytronics of 2D materials make them highly appealing for spintronics, particularly memory technologies like magneto-resistive random-access memory (MRAM) [97–100], which rely heavily on ultrathin materials and interfaces to provide the essential capabilities [9,101–103]. Despite the rapid advancement of research on 2D magnetic material but there is an elusion that requires a thorough study that outlines the manufacturing techniques, characteristics, and spintronic devices of 2D magnetic materials.

In this review article, we focused on the significance of 2D magnetic materials for diversity and challenges in the development of spintronic devices. This review's objective is to provide an up-to-date summary and analysis of the most current developments in the aforementioned study areas. We first describe the respective basics and manufacturing processes of 2D magnetic materials and then 2D magnetic material's physical characteristics and device applicability are comprehended. We have reviewed the structure and characteristics of 2D magnetic materials and summarized them in Table 3. We have utilized VESTA software [104–106] to draw different crystal structures of magnetic material and available structure files obtained from the materials project [107,108] and crystallography open database [109–112]. Finally, we highlighted the issues and concerns that will need to be overcome for further the study of 2D magnetic materials and their application in spintronics.

Development of materials

Since TMDCs and many 2D magnetic single crystals are not acknowledged as existing in nature, they must be manufactured in a laboratory. Van der Waal's (vdW) crystals may be easily obtained and structured in a nanoscale using chemical and mechanical methods [6,63,113–117]. Here, we discuss various

production techniques of 2D magnetic materials that are currently utilized, including prominent and cost-effective methods.

Mechanical exfoliation

Famous scientists named Geim and Novoselov developed this method in 2004 to produce graphene [118]. Due to its affordability and adaptability, mechanical exfoliation has become a facile approach for creating 2D materials. To get unique layers of 2D materials (from single to few layers) while maintaining crystal symmetry and other characteristics, mechanical exfoliation [119] is a simple method of material synthesis. Fig. 1(a) depicts the mechanical exfoliation process. In this technique, we employ regular scotch tape to exfoliate atomically thin layers of 2D materials from bulk crystal materials. A one-sided adhesive tape is used to exfoliate the 2D bulk material. When we press the scotch tape again on the surface of the substrate [120], the top layer turns into part of the tape. When the bottom layer peels away, it becomes a part of the substrate [120]. An optical image of mechanically exfoliated flake of $\text{Cr}_2\text{Ge}_2\text{Te}_6$ (CGT) is shown in Fig. 1(b). Multi-layer flake Fe_3GeTe_2 (FGT) is illustrated in Fig. 1(c). As of right now, mechanical exfoliation from a bulk single crystal is the primary method used to create atomically thin layers. Thin layers of prominent 2D magnetic substances such as CrI_3 [121], CrSiTe_3 [122], CGT [50], NiPS_3 [123], FePS_3 [122,124], Fe_3GeTe_2 [125,126], CrTe_2 [127] have been set up by the mechanical exfoliation process. The CGT and materials that are associated with group MnX_3 are stable as compared to other magnetic materials. Since, most 2D magnetic materials are typically unstable in the air, and that's why they are usually exfoliated from bulk materials using a glovebox. Thin flakes of FGT [128] and CrI_3 [129] oxidize so quickly in the air, that a glove box is recommended for experimentation. Scalable production of 2D flakes is highly challenging, however, mechanical exfoliation can prepare single layer 2D materials rather well. However, due to its smaller production, this approach is not suitable for large-scale practical applications.

Chemical vapor deposition (CVD)

The synthesis of 2D magnetic materials on a large scale has remained a significant challenge. The chemical vapor deposition (CVD) technology has received a great deal of interest since it allows us to produce 2D materials on a large wafer-scale, which has major implications for practical applications such as

large-scale integrated circuits [130]. The CVD process has been widely utilized to create thin film coatings of an extensive range of materials, including insulators, semiconductors, and metals [131]. This technology is utilized to not only synthesis continuous single films with defined thicknesses but also to directly build layered hetero-structures that avoid interfacial contaminants formed during the layer-by-layer transfer procedure. The vdW epitaxial growth has emerged as a key method for the controlled CVD development of 2D materials on a variety of substrates, including mica [132], gallium nitride [133], sapphire [134], and quartz [135]. A substantial study has been accomplished in developing catalytic substrates [136], choosing appropriate substrates [134], annealing and selective patterning of the substrates in recent years in order to attain a large domain size, rapid growth rate, and consistently oriented growth of 2D multiple materials [137,138]. Many various 2D magnetic materials were synthesized by CVD such as CrTe_2 [139], CrSe_2 [140], VS_2 [141,142] and Cr_5Te_8 [143].

We briefly described the CVD setup as shown in Fig. 2(a) where the CVD has been employed to produce high-quality Cr_5Te_8 [143] nanosheets on a newly sliced mica substrate. As tellurium (Te) and chromium (Cr) are precursors, respectively, Te powder (200–800 mg, Alfa, 99.999 %) and CrCl_2 powder (5 mg, Aladdin, 99.9 %) were utilized. Te powder was loaded in a quartz boat upstream of the hot zone, at temperature of 500 °C. At the tube furnace's heating core, a mica substrate was put face-down atop a quartz boat containing the CrCl_2 source. High-purity argon (Ar, 300 sccm) gas was injected to purge the reaction chamber for 10 min before the heating operation. Then, with a carrier gas combination of 50 sccm Ar and 20 sccm hydrogen (H_2), the furnace temperature was ramped up to the growth temperature (600–900 °C) and sustained there for 10 min to produce Cr_5Te_8 nanosheets. By using a hydrofluoric acid (HF)-assisted wet transfer technique, the as-synthesized Cr_5Te_8 nanosheets were transported onto the target substrate (such as SiO_2/Si (300 nm) substrate and copper grid). The findings show that the effective epitaxy of such nonlayered Cr_5Te_8 crystals down to the 2D limit depends on an atomically flat mica surface with a low migration barrier. By varying the growth temperature from 600 to 900 °C, the controlled synthesis of Cr_5Te_8 was tested. The statistical results (Fig. 2b) and representative optical pictures of as-grown Cr_5Te_8 nanosheets on mica show a regular progression of the thickness (from 1.2 to 26 nm), lateral dimension (from 5 to

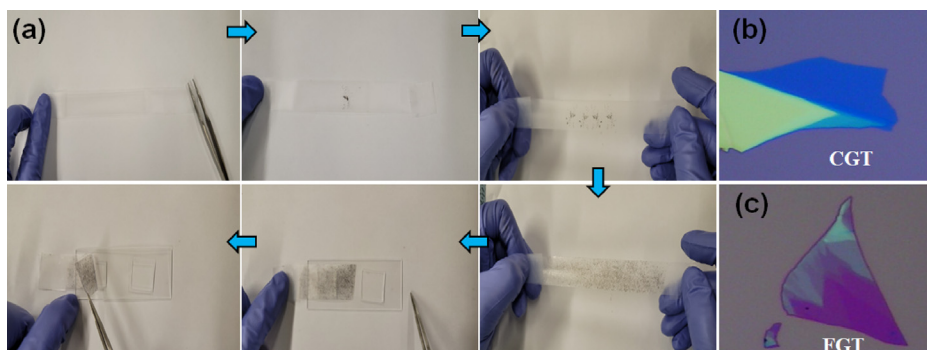


FIG. 1

(a) Process of mechanical exfoliation. (b) Optical image of a mechanically exfoliated flake of $\text{Cr}_2\text{Ge}_2\text{Te}_6$ (CGT). (c) Multilayer exfoliated flake of Fe_3GeTe_2 .

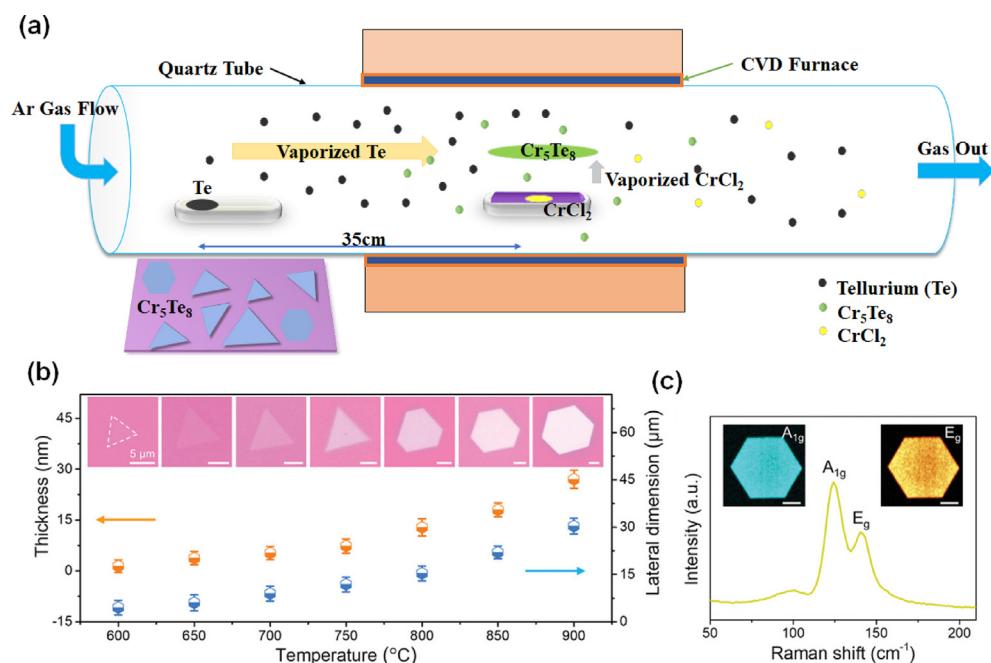


FIG. 2

(a) Schematic CVD setup for the growth of Cr₅Te₈. (b) Data of the lateral dimension and thickness of Cr₅Te₈ nanosheets grown at various temperatures. The accompanying typical optical images are inset. (c) As-grown Cr₅Te₈ nanosheet's Raman spectra. The Raman intensity mapping pictures for the E_g and A_{1g} modes are included as insets and every scale bar is 5 μm. Figures (b) & (c) is reused with the approval of reference [143], copyright, Advanced Materials 2022.

30 m), and even shape (from triangle to hexagon) with an increasing growth temperature. Moreover, the thickness of CVD-grown Cr₅Te₈ nanosheets may be determined using the Raman intensity and frequency characteristics of Cr₅Te₈ nanosheets, both of which exhibit considerable thickness dependency. The out-of-plane A_{1g} and in-plane E_g vibration modes, respectively, are represented by the two principal distinctive peaks in the Raman spectra of the Cr₅Te₈ nanosheets (Fig. 2c), which are located at 124 cm⁻¹ and 142 cm⁻¹. Additionally, the very consistent thickness and phase structure of Cr₅Te₈ nanosheets are seen in Raman intensity mapping pictures of the A_{1g} and E_g modes [143]. The remarkable and air-stable FM feature of Cr₅Te₈ together with its controlled synthesis offers up new avenues for the exploration of 2D magnetism and spintronic device applications. Therefore, we would like to address how the substrate could affect the growth process in detail. Firstly, mica is a common substrate for the growth of 2D composite materials. For instance, Ji *et al.* [132] synthesized monolayer MoS₂ using low-pressure CVD on an inert and nearly lattice-matched mica substrate. But fluorophlogopite mica is appropriate for the epitaxial growth of other 2D compound materials such as CrSe [144] which is attributed to its flexibility, surface inertness, and atomic flatness. Secondly, to make catalytically active substrates, one can lower the substrate's active surface energy, which will facilitate the initiation and development of 2D materials during CVD growth. Typically, sapphire, SiO₂, and other substrates with low catalytic activity are utilized, which results in slow development and difficulties in using growth behavior control to provide a monolayer feature. The third technique involves pre-treating the substrates to provide a certain dominating crystal plane for the epitaxial development

of 2D materials, resulting in 2D domains with a predictable orientation. In general, optimized substrates are crucial to the epitaxial growth of 2D materials.

Molecular beam epitaxy (MBE)

Large-scale 2D FM thin film production is still a challenge despite experimental advancements. So far, chemical vapor transfer (CVT) has been the primary method used to create ferromagnetic single crystals like Fe₃GeTe₂ and CrGeTe₃ [145,146]. These materials are easily exfoliated by the simple exfoliation approach because of the vdW's contact between layers, but the resulting nano-devices are mostly constrained by the size of the sample [147]. Consequently, the development of large-scale 2D FM thin films successfully develops a focus since it not only promotes physics research at a quick pace but also offers the chance to study diverse heterostructures [148]. MBE is a technique used to grow thin films of crystalline materials, such as semiconductors and metals, with extremely precise control over their thickness and composition at the atomic scale [149]. MBE is widely used in the semiconductor industry to create the layers of materials used in electronic devices. In this process, a beam of molecules and atoms in an ultra-high vacuum environment is kept incident onto a heated substrate. The layers are incredible because their composition may be rapidly altered, resulting in crystalline surfaces that are roughly atomically abrupt. One of the key benefits of MBE is its capability to accurately manage the thickness and composition of the deposited film. This is achieved by carefully controlling the temperature and pressure of the deposition chamber, as well as the flux of atoms or molecules from the source material. As a result, MBE can produce films with atomic-level precision and uniformity, making it a

powerful tool for research and development in materials science. Overall, MBE is a versatile and powerful technique for the precise growth of high-quality 2D materials and thin films and is widely used in both research and industrial applications [150–152].

Sample growth by MBE

A commercial Veeco 930 molecular beam epitaxy (MBE) system was used to grow Cr_2Se_3 (CS) and Cr_2Te_3 (CT) samples [153]. The sample was heated by a tungsten filament behind the molybdenum plate holder, as shown in Fig. 3(a). A metal plate held a quarter of a 2-inch single-sided polished sapphire. Each source was sublimated using a Veeco solid source effusion cell: an 85 cc PBN crucible on an EPI-85-LT cell for Se (99.999+%, Alfa Aesar) and Te (99.999 %, Sigma-Aldrich), and a 10 cc W crucible with a 10-cc alumina liner on an EPI930-10-HT cell for Cr (99.999 %, Alfa Aesar). An additional ion gauge was utilized to test the beam equivalent pressure flux ratio for Se, Te, and Cr, which was kept constant at around 1:50 (Cr: Se) for CS growth at 580 °C and 1:20 (Cr: Te) for CT growth at 320 °C. After growth, the sample was annealed for 30 min at 320 °C. A photograph of an as-grown sample on the sample holder is shown in Fig. 3(b). Fig. 3(c) represents the image of elemental mapping region of CT/CS heterojunction (20 nm/20 nm) deposited on the sapphire substrate and Al_2O_3 is utilized here as capping layer. Fig. 3(d) highlights the HR-TEM image that denotes the growth of heterostructure of along c -axis. The images describe the alternating sequence of in-plane Se (Te). These layers had quasi-two-dimensional (2D) properties because the Cr atoms were bound to Se (Te) atoms that were nearby.

Furthermore, we have reviewed the MBE-grown structure of Fe_3GeTe_2 (FGT). In a high-vacuum MBE technique, FGT films with excellent crystalline quality were produced on GaAs [70]. More HRTEM studies were conducted to identify the FGT film's layered structure on (1 1 1) GaAs (Fig. 3e). The associated marked area electron diffraction (SAED) collected from the interface region is shown in Fig. 3(f), and it can be displayed that the FGT film and GaAs have the crystallographic correlation like as {0001} Fe_3GeTe_2 /[1 1 1]GaAs. The elemental distribution of Fe, Ge, and Te in the film is checked by energy-dispersive X-ray spectroscopy (EDS) maps acquired from the film area, as shown in Fig. 3(g). The FGT and CT/CS thin films are seen to grow evenly on the substrate and exhibit a high-crystalline characteristic after all these characterizations [70,153].

Proximity effect

In magnetic materials, the proximity effect refers to a phenomenon that occurs at the junction between a metal/superconductor and a magnetic material. When a metal/superconductor is brought in close proximity to a magnetic material, the magnetic properties of the material can influence the behavior of the metal/superconductor [154,155]. The proximity effect in magnetized superconductors is a result of the exchange of quasiparticles between the superconductor and the magnetic material. Quasiparticles are a type of particle-like excitation that arises in condensed matter systems, and their exchange can result in the modification of the superconducting properties of the material. A recent method of systematically designing materials is suggested by improvements in the scaling down of heterostructures,

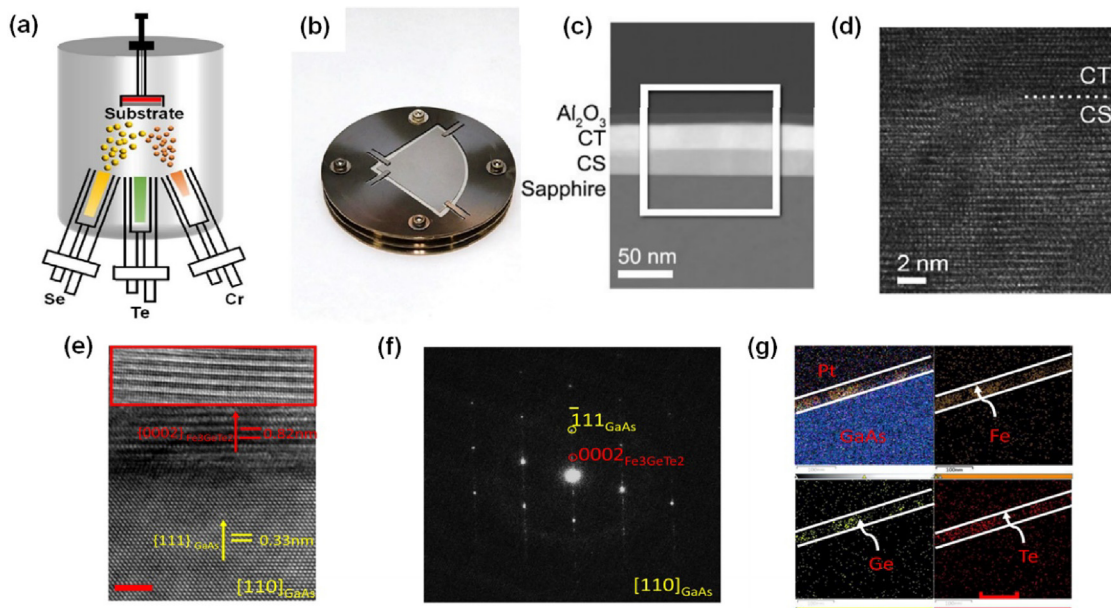


FIG. 3

(a) The MBE growth chamber is depicted schematically. (b) A photograph of a quarter of a 2-inch sapphire wafer after the growth process. (c) Cross-sectional EDX mapping and line-profiles, CT/CS heterostructure mapped by EDX (highlighted by white square). (d) Cross-sectional HR-TEM for CT/CS. The layer of protection was made of PMMA. The grain boundary of rotating domains can be observed with white dots. This figure is replicated with the approval of the reference [153], Copyright ACS Nano. (e) Fe_3GeTe_2 thin film formed on (1 1 1) GaAs was the subject of cross-section HRTEM pictures, The Fe_3GeTe_2 region's top inset, represented by a red rectangle, demonstrates that its layer distance is 0.82 nm. (f) Corresponding SAED. (g) Results of the EDS mapping show that Fe, Ge, and Te atoms are distributed uniformly throughout the film in the area between the outlined white lines. This figure is reproduced with the consent of reference [70], Copyright, npj 2D Materials and Applications.

increased interface quality, and atomically thin two-dimensional materials. Material can change by proximity effects, wherein it takes on the characteristics of its surroundings, such as superconductivity, magnetism, topological non-triviality, or improved SOC. Such types of effects not only overcome the numerous restrictions of the traditional doping or functionalization approaches of material design but also serve as a supplement to them. The characteristics that do not exist in any essential region of the under-consideration heterostructure can be realized in proximities materials. While magnetic and spin-orbit proximity effects and their uses in spintronics are the main emphasis, the presented concepts also offer a wider framework for utilizing additional proximity effects to modify materials and recognize innovative mechanisms. Recent advances in proximity effects, which may change a given material's characteristics through its nearby areas to develop magnetic or superconducting properties have shown a fundamentally new method for tailoring materials. The proximity effect in magnetic materials has important implications for the design and performance of superconducting devices, such as superconducting quantum interference devices (SQUIDs) and superconducting magnetic energy storage (SMES) systems. By understanding and controlling the proximity effect, researchers can optimize the performance of these devices and develop new technologies based on superconductivity and magnetism. In this overview, we present a much more comprehensive depiction of this effect as a general method for transforming a large class of materials that can surmount the intrinsic constraints of doping and functionalization. As graphically shown in Fig. 4(a), prospects to create proximities materials already exist at equilibrium. As seen in Fig. 4(b), superconducting/Metallic qualities can permeate from a superconductor onto an adjacent normal metal area that would not be superconducting/metallic on its own. Correspondingly, in magnetic proximity effects, a ferromagnetic (FM) magnetization penetrates an

adjacent nonmagnetic (NM) zone. Surprisingly, superconducting proximity effects may achieve orders of magnitude greater lengths than other proximity effects, even exceeding $> 100 \mu\text{m}$ in pure metals at sub-Kelvin temperatures [156,157]. The procedure of Andreev's reflection is what causes superconducting proximity: at the junction with a superconductor, an incoming electron is *retro*-reflected as a hole, along with the generation of a Cooper pair [88]. While proximity effects are usually well-defined as the exchange of an ordered state, such as magnetism or superconductivity to an area where it was formerly absent without knowingly altering its electronic structure, this thing has recently been expanded to refer to proximity-induced topological properties or SOC [158,159].

In bulk materials, the size of sample frequently surpasses the typical lengths of proximity effects, permitting them to be overlooked. Nevertheless, in atomically thin vdW materials like h-BN, graphene and TMDCs [160–163], just as short-range magnetic proximity effects transcend their thickness and significantly alter the transport and optical characteristics. For instance, pristine graphene has zero bandgap and is massless, with a linear dispersion around the K-point in the Brillouin zone (Dirac cone), a minor SOC, and a spin unpolarized density of states (DOS) [164,165]. However, proximity effects from nearby materials significantly change the nature of graphene, allowing it to acquire either negative or positive effective mass [166], SOC [155,167], spin polarization [168] or even superconductivity [169,170]. One of several vdW materials that exemplify the new tendencies in modifying characteristics through proximity effects is graphene. Additionally, as nanostructures get smaller and interface quality gets better, other kinds of materials can also be used as a platform to show proximity effects. The proximity effects can be utilized to create exotic topological phases that replicate large-scale characteristics of heterostructures that are insensitive to disorder and local perturbations, take the lead to

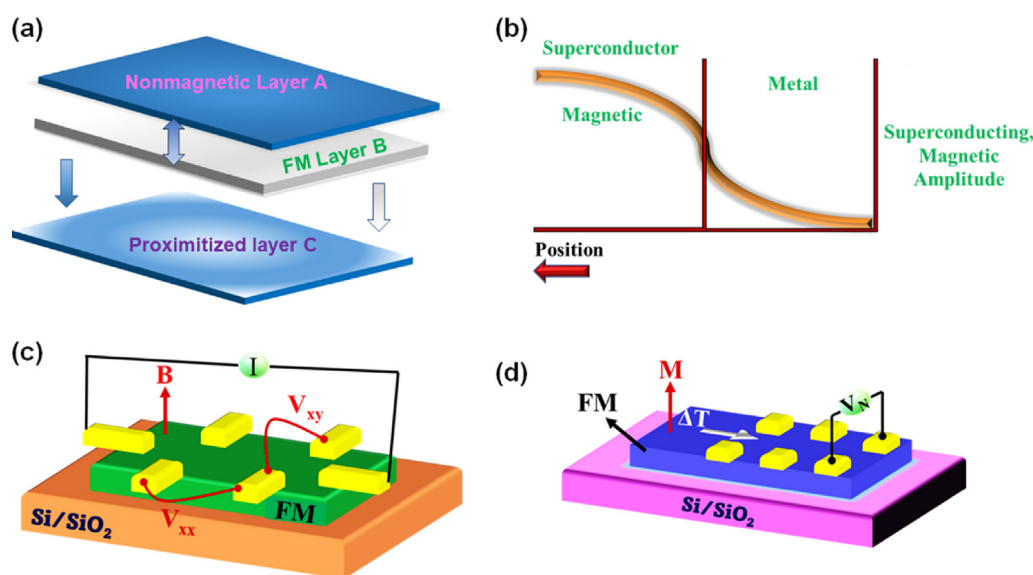


FIG. 4

(a) Proximity-modified layer C with the effect of the corresponding FM: layer B and layer A. (b) Superconductivity/magnetic effect crossing a boundary and entering a normal (non-superconducting, nonmagnetic metal) material/region. (c) A schematic diagram illustrates the measurement approach of the ANE. (d) Schematic shows the measurement setup of Anomalous Nernst effect (ANE).

purposes such as ultra-high density magnetic storage using magnetic skyrmions [171–173] or topologically locked quantum computing with non-Abelian quasiparticles [174,175]. The potential low-power topological quantum computing and electronics/spintronics may benefit from the spin-polarized surface electronic states of 3D topological insulators (TI), which exhibit unique physical features [176]. One of the distinguishing features of magnetization perpendicular to the plane caused by proximity coupling due to the ferromagnetic (FM) layer on the TI is the production of anomalous Hall conductance in the gapped surface state [177–180]. Additionally, the spin accumulation could make it possible to control electrically the TI surface ferromagnetism that results from the FM proximity coupling at the FM-layer/TI interface. The current at the TI surface state is primarily shunted by the FM-metal layer in the widely investigated FM-metal/TI devices. As a result, substituting an FM insulator (FMI) for the FM metal would be effective in controlling magnetization with reduced electric current [181–183], which is also necessary to detect quantum anomalous Hall states with non-trivial Chern numbers [180]. Since magnetically doped TI [184,185] and FMI/TI heterogeneity [181] have shown that the magnetization in the FMI layer may be partially altered by current excitation, the switching ratio of the magnetization has been constrained to 50 %. Such a low ratio, which is likely due to the non-homogeneous nature of doping-induced ferromagnetism and weak magnetic coupling at the interface of FMI/TI, leads to the generation of multidomain states after each switching operation and makes topological invariant control problematic. Thus, it is envisaged that selecting an appropriate FMI with extraordinary crystallinity and better interfacial coupling, thereby, results in a high switching ratio [186].

Proximity effect stimulated anomalous Hall effect (AHE) in CGT/BST

When a magnetic field is engaged to a metal through which a current pass, the Hall transverse electric field is created when current carriers are bent in the field. The imbedded magnetic moments in a ferromagnetic metal cause an AHE [187]. The AHE is caused by the fact that electrons are preferentially deflected in one direction when a current flows through a magnetic substance. This generates an extra current perpendicular to the driving current, which disappears if the material is non-magnetic [188]. The AHE has developed a significant tentative tool for physicists since it is dependent on both the electronic and magnetic characteristics of the material. Energy-efficient spintronics innovation may depend on the electrical control of magnetization. The schematic shows the AHE measurement approach in Fig. 4(c). While a temperature gradient is utilized to a magnetic material, a voltage is generated that is perpendicular to both the magnetization and flow of heat is referred as anomalous Nernst effect (ANE) [189–192] and the measurement setup for ANE is shown in Fig. 4(d), which was long supposed to be proportional to the magnetization value. However, in general, ANE is thought to be caused by a net Berry curvature of all bands at the Fermi level. Many traditional ferromagnets display a succession of magnetization-reliant transverse transport mechanisms as linear responses to a temperature gradient or electric field, several of which are intricately connected.

The surface magnetic state of a magnetic topological insulator with a magnetically gapped surface state and spin-polarized electrons may be electrically controlled in addition to exhibiting exotic topological phases important to the quantum anomalous Hall state. In the hetero-junction of topological insulator ($\text{Bi}_{1-x}\text{Sb}_x$)₂Te₃ (BST) and the FM insulator Cr₂Ge₂Te₆ (CGT), efficient current-stimulated switching of the surface ferromagnetism was revealed [193], and the proximity-stimulated ferromagnetic surface states perform two vital roles: efficient charge-to-spin current exchange and the development of large AHE. The recent findings may help with the electrical regulation of dissipationless topological-current circuits. Here, we reviewed the provision relating magnetization switching in an intrinsic FMI (CGT)/TI heterostructure based entirely on telluride as shown in Fig. 5(a). Recent research has revealed that the layered FMI CGT [186,194] exhibits substantial proximity coupling with the TI surface state, as demonstrated by the detection of a significant AHE at the exchange gap established in the proximity-coupled surface state [195]. Molecular beam epitaxy-based (CGT/BST) bilayer films on substrates of InP (111) (Fig. 5a) is explored. Hall bars with a 10 μm width for the electrical transport tests and magnetization switching are illustrated in Fig. 5(b). The magnetization of CGT (12 nm)/BST (6 nm) structure at $T = 2\text{ K}$ (Curie temperature: $T_C = 80\text{ K}$) reveals the hysteresis loop with the out-of-plane easy axis, which is almost similar with the characteristics of CGT monolayer itself [194] as shown in Fig. 5(c). For the CGT (12 nm), BST ($x = 0.5$, 6 nm), and CGT (12 nm)/BST ($x = 0.5$, 6 nm) bilayer films, the T dependency of sheet resistance R_{xx} is shown in Fig. 5(d). The resistance of heterostructure film is equivalent to that of the BST monolayer film, but the CGT monolayer film displays an extremely high insulating characteristic ($>1\text{ M}\Omega < 50\text{ K}$). We first go through how the AHE for various samples of the CGT/BST ($x = 0.5$, 6 nm) bilayers depends on the CGT layer thickness (t_{CGT}). The Hall resistance R_{yx} of several samples with similar AHE responses is shown in Fig. 5(e). Additionally, the magnetization hysteresis curve of the CGT layer ($t_{\text{CGT}} = 12\text{ nm}$; Fig. 5(c)) and the coercive field of the hysteresis loops are also stable. At zero magnetic field, the value of the anomalous Hall resistance R_{yx}^{AH} (250 Ω) is essentially constant regardless of t_{CGT} ($2.9 \leq t_{\text{CGT}} \leq 12\text{ nm}$), demonstrating that the Berry curvature produced at the magnetically gapped TI surface state. The CGT (2.9 nm)/BST ($x = 0.5$, 6 nm) bilayer's current-induced magnetization switching is shown in Fig. 5(f), [197]. Current pulses (J_x) were incorporated with in-plane magnetic fields ($\mu_0 H_x = \pm 0.1\text{ T}$), which were substantially less than the anisotropy fields ($H_K \approx 0.9\text{ T}$; to execute the perpendicular magnetization switching [196–198]. The R_{yx} was measured after each current pulse injection with a significantly lower probe current of $J_x = 10\text{ }\mu\text{A}$. As seen in Fig. 5(f), the sign of the Hall resistance is reversed when the injection current pulse's amplitude surpasses the switching threshold current $J_x^{\text{sw}} \approx 2.0\text{ mA}$. The switching polarity is reversed when the in-plane magnetic field is reversed, as predicted by the anti-damping-like spin-orbit torque switching [196,197] (H_x (blue curve) $\rightarrow -H_x$ (red curve)). The remnant Hall resistance $R_{yx}^{\text{AH}} = 228\text{ }\Omega$, where the ratio $R_{yx}^{\text{sw}}/R_{yx}^{\text{AH}} \approx 0.88$ concerned with the reversed magnetization ratio in the CGT, is equivalent to the alteration of Hall resistance $R_{yx}^{\text{sw}} = |R_{yx}(J_x \rightarrow +0, H_x < 0) - R_{yx}(J_x \rightarrow 0, H_x < 0)|/2 = 200\text{ }\Omega$. This high number indi-

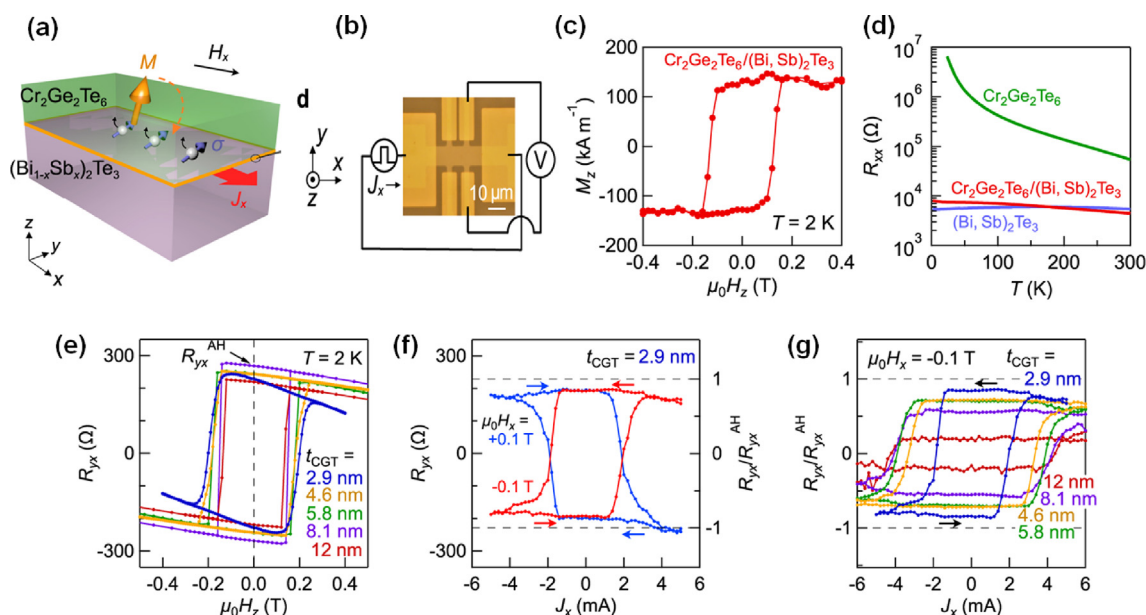


FIG. 5

(a) Schematic representation of magnetization switching of bilayer materials CGT/BST. M in CGT is overturned by insertion of current along x-direction (J_x) when subjected to in-plane magnetic field (H_x). (b) Image of a Hall bar device under an optical microscope showing the measuring setup. (c) Reliance of the magnetization M_z in the CGT (12 nm)/BST ($x = 0.5$, 6 nm) bilayer at 2 K with respect to the out-of-plane magnetic field ($\mu_0 H_z$). (d) Temperature (T) dependent longitudinal resistance R_{xx} for the single-layer CGT (12 nm; green), single-layer BST ($x = 0.5$; 6 nm; purple), and bilayer CGT/BST (12 nm/ $x = 0.5$, 6 nm; red) films. (e) Hall resistance (R_{yx}) in CGT/BST ($x = 0.5$, 6 nm) bilayer devices at 2 K with different CGT thicknesses ($t_{\text{CGT}} = 2.9$, 4.6, 5.8, and 12 nm) as a function of magnetic field. (f) Magnetization switching with current pulses J_x with in-plane applied magnetic field ($H_x = \pm 0.1$ T). (g) Magnetization Switching for various thicknesses (t_{CGT}) of CGT. This figure is re-used by the agreement of reference [193], exclusive rights, Nature communication 2021.

cates that practically entire magnetization switching has occurred, leading in the switching of FM surface states and topological spin configurations in the surface state. After establishing current-induced magnetization switching, it was investigated the t_{CGT} dependency of J_x^{sw} in order to calculate the efficiency switching ratio. As demonstrated in Fig. 5(g), as t_{CGT} grows, the J_x^{sw} increases, whereas the switching volume fraction evaluated by $R_{yx}^{\text{sw}}/R_{yx}^{\text{AH}}$ drops. These findings show that the effective magnetization switching, and the massive proximity induced AHE are compatible, opening the door to electrically manipulating topological quantum states. This opens a new path for previously unheard-of control of dissipation-less topological current devices by connecting the spintronic functions to the topological quantum states.

Furthermore, we also looked at the proximity effect in heterostructure made of monolayer WSe_2 and bi/trilayer antiferromagnetic (AFM) CrI_3 . It was demonstrated that the interfacial CrI_3 layer dominates the spin-dependent charge transmission among WSe_2 and CrI_3 , whereas the proximity exchange field is extremely sensitive to the layered magnetic configuration. This study states a van der Waals engineering approach to manage proximity effects and investigate junction magnetic order [199]. The magnetic order can have a substantial influence on the non-magnetic component's characteristics at the junction produced by non-magnetic and magnetic material [200], which can reveal novel functions not present in the separate materials [77]. Additionally, it was demonstrated that using the spin-valley characteristics of a monolayer semiconductor, it was possible to detect multilayer AFM domains within a vdW magnet, which

would be difficult using traditional magnetometry methods. In conclusion, this study makes use of the many magnetic configurations available in layered AFM (CrI_3) to shed light on the magnetic proximity effects of a vdW magnetic heterojunction. Having this knowledge will be crucial for creating magnetic vdW materials based devices [201].

Thermoelectric phenomenon due to proximity effect

Proximity effect stimulated anomalous Nernst effect (ANE) in CGT/ WTe_2
ANE and AHE in semiconductors and metals, respectively, are associated by the Mott relation [202–204] via the energy derivative of the anomalous Hall conductivity, and their corresponding unquantized coefficients are quantitatively defined by either intrinsic or extrinsic strategies. We have summarized the ANE mechanism for various materials in Table 1. In the heterostructures containing a ferromagnet, ferromagnetic order may often be created by doping or interfacial proximity coupling. The latter method has been shown in a collection of material systems, consisting of graphene [205] and topological insulators [206,207]. The heterostructure technique is particularly appealing for ML- WTe_2 since vdW heterostructures such as $\text{CrI}_3/\text{WSe}_2$ [201] and $\text{CrI}_3/\text{WTe}_2$ [208] have proved to be extremely successful in producing proximity coupling due to their atomically flat surfaces. The tailoring of specific connections at the atomically flat interfaces between different layered materials and the induction of novel physical facts are both very flexible options provided by van der Waals heterostructures. Monolayer 1 T' WTe_2 and insulating ferromagnet few-layer $\text{Cr}_2\text{Ge}_2\text{Te}_6$ into close contact in a heterostructure, an introduction of ferromagnetic order in the

TABLE 1

Summary of different magnetic materials, for Anomalous Nernst effect (ANE).

Sr No	Material	Type/Magnetic field Alignment	Curie Temperature (T_c)	Anomalous Nernst Voltage (V_{ANE})	Anomalous Nernst coefficient (S_{ANE})	Reference
1.	$Mn_5Ge_3C_{0.8}/Ge$	FM Semiconductor $\perp$	~ 450 K	–	$2 \mu V K^{-1}$	[211]
2.	Ni/Pt	FM Metal $\perp$	627 K	$2 \mu V$	$1 \mu V K^{-1}$	[212]
3.	$Co_x(MgO)_{1-x}$	FM Composite $\perp$	110 K	$2 \mu V$	$0.4 \mu V K^{-1}$	[213]
4.	Co_2MnGa/AlN	FM Composite $\perp$	101 K	1.5	$4.9 \mu V K^{-1}$	[214]
5.	Co_2MnGa/Si	FM $\perp$	1015 K	–	$82.3 \mu V K^{-1}$	[215]
6.	Fe_3GeTe_2	FM $\perp$	270 K	$2 \mu V$	$0.3 \mu V K^{-1}$	[192]
7.	CGT/ WTe_2	FM Insulator $\perp$	61 K	$1 \mu V$	–	[209]

former via the interfacial exchange interaction is reviewed here [209]. The ANE, AHE, and anisotropic magnetoresistance effects are manifestations of the ferromagnetism in WTe_2 . The ML- WTe_2 /CGT vdW heterostructure is depicted schematically in Fig. 6(a). The CGT is an FM insulator under its Curie temperature T_c of 61 K with magnetic anisotropy perpendicular to its atomic layers [50,145,210]. Fig. 6(b) depicts the ANE device's structural layout. Fig. 6(c) is the optical image of the device earlier to transfer of ML- WTe_2 /CGT composite layer. A heater was designed primarily to provide a horizontal temperature gradient ∇T perpendicular to the magnetic field and the voltage channel to assess ANE responses. The second harmonic voltage recognition from electrode pairs 3–8, 4–7, and 4–5 was measured (as indicated in Fig. 6(c)). Since, an out-of-plane magnetic field H_z was being swept while an alternating current (AC) was being run

through the heater (through electrode pair 1–2 in Fig. 6(c), at 4 K temperature. Fig. 6(d) highlights the H_z -dependency of the voltage along three electrode sets. All three channels exhibit hysteresis behavior, which is like the anomalous Hall loop in CGT/Pt [210], apart from where the tilted loops are almost closed. We have seen that the CGT flake thickness affects the structure of the hysteresis loop and that it can totally collapse over a particular thickness due to increased dipolar interaction. A similar logic applies to the SOC source of the AHE in ferromagnets [204], since the conventional Hall effect cannot account for the enormous size of the Hall hysteresis loops if they were created by a stray field. To further study the relationship among magnetic order in CGT and stimulated ferromagnetism in ML- WTe_2 , the authors measured V_{ANE} temperature dependency. At various temperatures, Fig. 6(e) depicts the looping of normalized ANE voltage

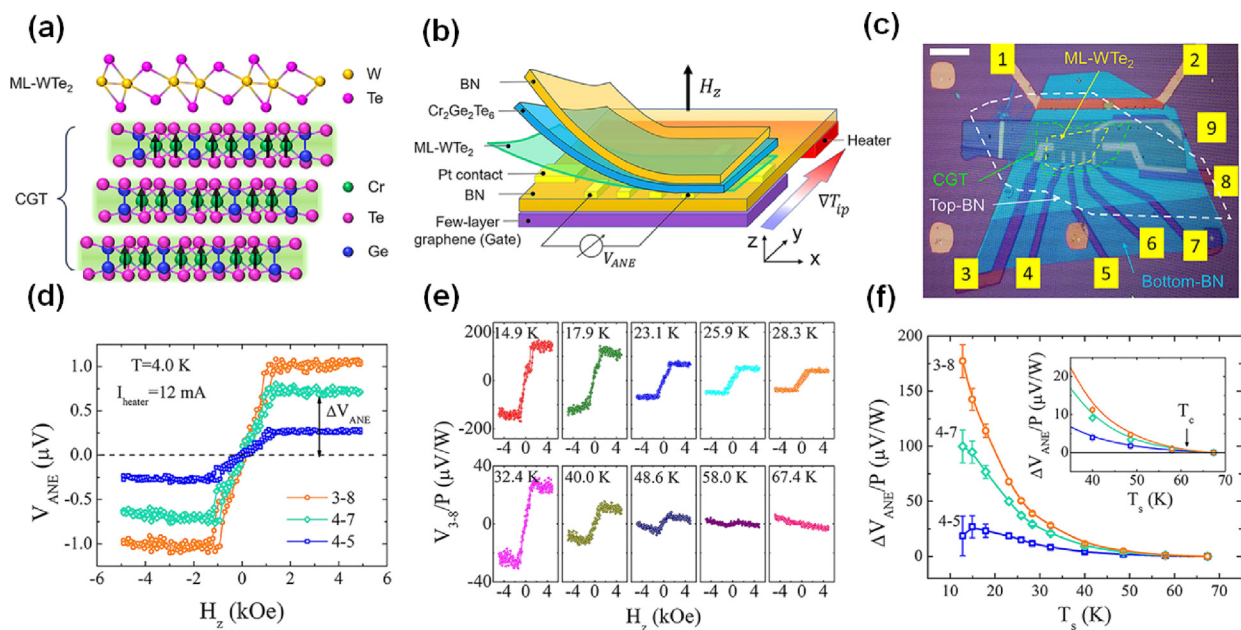


FIG. 6

(a) Schematic illustration of WTe_2 /CGT heterostructure (bilayer). (b) Schematic representation of ANE device configuration. (c) Optical image of designed device for ANE phenomenon. (d) Comparison of measured ANE voltage at various electrodes pairs. (e) Representation of ANE signal at various temperatures from the electron pairs 3–8. (f) ANE voltage signals as a function of temperature from different electrode pairs as mentioned in the optical image. This figure is reused with the consent of reference [209], exclusive rights, Nature Communication, 2022.

by power P , V_{ANE}/P . As of the slanted loop form, the authors calculated the saturation values of V_{ANE}/P on negative and positive fields and plotted the half difference in Fig. 6(f) for the three pairings. They all disappear when the device heats up and the sample temperature T_s approaches 61 K, the T_c of CGT. The calibrated WTe_2 sample resistance is used as a sensitive thermometer to calculate T_s . ANE is obviously linked to the ferromagnetic order parameter of CGT. These findings from several electrodes show that the magnetized quantum spin Hall insulator has spin-polarized edge states [209].

Spin transport phenomena in TMDC

Non-linear anomalous Hall effect (AHE) in TMDC material

Certain time-reversal symmetric systems, such as magnetic materials in the case of the AHE and materials exposed to an external magnetic field in the case of the conventional Hall effect, exhibit AHE. Here we have reviewed the AHE in tungsten ditelluride (WTe_2). The observed result may be explained by the bias current induction of an AHE, which causes an out-of-plane magnetization. The Hall conductivity's temperature dependency further implies that the nonlinear AHE seen is caused by both extrinsic spin-dependent scatterings and intrinsic Berry curvature dipoles. In these observations, numerous WTe_2 -based samples with different thicknesses were evaluated. The Hall bar device with aligned electrodes is illustrated in Fig. 7(a). The outcome of a conventional Hall bar device's fundamental characterization is shown in Fig. 7(b, c). Below around 100 K, with a residual resistivity $\rho_{\parallel}$ of about $50 \mu\Omega\text{cm}$, the temperature reliance of the longitudinal resistivity alongside the crystal a-axis (Fig. 7(c)) exhibits typical metallic behavior. When the temperature rises over 100 K, the dependency departs from a linear relationship,

which is likely a sign of non-degenerate electron gases [216] (in semi-metallic WTe_2 , the low densities allow for simple thermal excitation of the electron and hole gases). The transverse (Hall) resistivity $\rho_{\perp}$ (red line) at 1.8 K and the magnetic-field dependency of $\rho_{\parallel}$ are shown in Fig. 7(c). The field is out of plane, the significant non-saturating $\rho_{\parallel}$ signal up to 14 T implies that the electron and hole densities are almost equally distributed [216–218]. Using the two-carrier analysis of $\rho_{\parallel}$ and $\rho_{\perp}$ reliance on magnetic fields. The carrier densities were calculated about $1.26 \times 10^{13} \text{ cm}^{-2}$ (electron) and about $1.16 \times 10^{13} \text{ cm}^{-2}$ (hole), and mobilities, about $1580 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ (electron) and about $960 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ (hole). These findings are consistent with the reported findings [216,219]. This work demonstrates that intrinsic Berry curvature dipole and extrinsic spin-dependent scatterings both lead to nonlinear Hall conductance in few-layer WTe_2 . These findings suggest that non-linear electrical transport can be used to investigate topological effects in solids [220–222], as well as applications in spin-orbit torque devices [223,224].

Non-saturated magnetoresistance in WTe_2

Magnetoresistance is a material's benchmark to alter the value of its electrical resistance in the existence of an applied magnetic field from the outside. Materials with high magnetoresistance are used as, magnetic memory [98,225], magnetic sensors [226,227], and hard drives [228], 2D magnetic materials have stimulated several important investigations in materials physics. Here, we highlighted the measurements of exceptionally high positive MR at low temperatures in TMDC materials such as WTe_2 : 452700 % at 4.5 K in a 14.7-tesla magnetic field and 13 million % at 0.53 K in a 60-tesla magnetic field. In distinction to certain other materials, even at extremely high magnetic

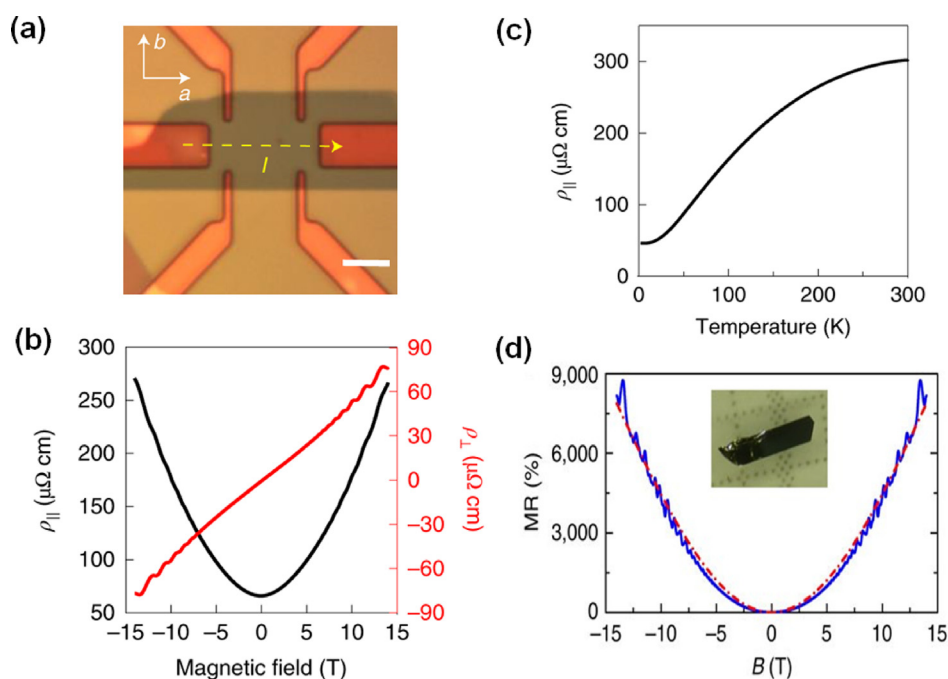


FIG. 7

(a) Optical image of the device which shows the Hall bars. (b) $\rho_{\parallel}$ with respect to the temperature. (c) $\rho_{\parallel}$ and $\rho_{\perp}$ with respect to the magnetic field at 1.8 K. This figure is replicated with the permission of reference [233], *Nature materials*, 2019. (d) MR of bulk- WTe_2 indicating non-saturating trend as a function of applied magnetic field (quadratic field dependence). This figure is reproduced by reference [232], copyright, *Nature communication*.

fields, the magnetoresistance value does not reach up to saturation in WTe_2 [217]. WTe_2 is very special since the substance is layered, and the system parameters may be altered by modifying the thickness with the use of a fast-exfoliating procedure. Fig. 7(d) demonstrates that the relative MR has a high, non-saturating quadratic dependency on magnetic field ($\mathbf{B}$), as described in prior studies [217,229–231]. In the study [232], the longitudinal resistance of multilayered WTe_2 and the unusual magnetic field dependency of the Hall resistance are quantitatively recreated by a traditional two-band model, nevertheless, a crossover to an Anderson insulator occurs for thinner crystals. These findings not only explain the cause of WTe_2 's magnetoresistance but also fully validate the two-band electron-hole transport hypothesis and show that atomically thin WTe_2 layers continue to be gapless semimetals [232].

Applications based on various 2D magnetic materials

The extraordinary potential of 2D magnetic materials makes them attractive for use in future spintronic devices. The efficiency with which 2D magnetic material's magnetization may be manipulated electrically is one of fascinating aspects. Electronics and spintronics are generally distinct from one another and spintronics explores the electron spin as an extra degree of freedom, resulting in excellent data storage, data transport, and computing efficiency. Spintronic-based devices such as spin-FET, spin valves/MTJ, and spin-orbit torques, have all been successfully built using 2D magnetic materials. We will primarily concentrate on the spintronic device presentations of 2D magnetic materials in this section.

Spin field effect transistor (Spin-FET)

In distinction to conventional transistors, a spin transistor could be able to provide non-volatile data storage in addition to better performance. Spin transistors have been developed using a variety of methods, but their development still has several issues. Datta and Das first proposed the notion of spin FET in 1990 [234]. The spin-FET is a three-terminal device that functions by regulating a spin-polarized current that flows from drain to source terminals [234]. In contrast to conventional transistors, a spin transistor could be able to provide non-volatile data storage in addition to better performance [129]. In Fig. 8(a), a typical spin FET schematic design is demonstrated. The source-drain terminals consist of FM material which causes the polarizing the spin of carriers that are introduced into the source, the FM source serves as a spin polarizer as demonstrated in Fig. 8(b). The FM drain functions as a spin filter, permitting electrons with a single polarization to enter whereas rejecting electrons with various polarizations as shown in Fig. 8(c). In between the FM source-drain, there is a channel medium that is normally semiconducting. There is a back gate that shows the metallic nature with an oxide layer separating it. Spin FET is anticipated to be a 2D spintronic chip with high processing speed and low power consumption. Gong *et al.* recently presented a spin FET that can be made with 2D 2H-VSe₂ [235]. But numerous latest investigations, however, indicate that VSe₂ is not a magnetic substance [236,237]. The spin tunneling FET (spin-TFET) has been experimentally proven to be an improvement over spin FETs. A spin TFET based on

a dual-gated graphene/ CrI_3 /graphene configuration was also recently explored by Jiang *et al.* [129].

New operating theories for spin transistors may result from the latest finding of 2D magnetic insulators with electrically switchable magnetic order and a powerful spin filtering effect, such as chromium triiodide (CrI_3). Here, we reviewed dual-gated graphene/ CrI_3 /graphene spin TFETs. The magnetic order in the CrI_3 tunnel barrier affects the device's ambipolar behavior and tunnel conductance. Unlike traditional spin transistors, such devices utilize the spin filtering effect (SPE) [238–240] to inject and recognize spin and electrical switching of the magnetization arrangement [61,62] in the tunnel barrier to regulate the transistor outputs instead of electrical control of spin current in the channel. This device showed a high-to-low conductance ratio of around 400 %. The schematics of the device's bottom and top gates are represented in Fig. 8(d). Two gates were made of graphene and h-BN (20 nm). Here, CrI_3 of various layers acted as a tunnel barrier. The schematics showed bilayer CrI_3 acted as a barrier with two magnetization structures: two ferromagnetically coupled layers (right) and an antiferromagnetically coupled layer (left). The gate can modulate the Fermi level of the graphene, which modifies the conduction of vertical structure in this device through resonant tunneling [241,242]. More vitally, the gate can also stimulate a spin-flip alteration in CrI_3 [61,62], which produces a substantial tuning in the tunnel conductance by the SPE. The spin filtering mechanism has been represented in the term of high tunnel magnetoresistance (TMR) in the device based on 2D CrI_3 [238,243]. We reviewed that the device characterizations were attained in cryostat down to 4 K, by biasing CrI_3 with graphene source and drain contacts, the tunnel conductance of the graphene/ CrI_3 configuration was determined. Fig. 8(e) demonstrates the gate dependency of conductance (G) uniform magnetic field (0.76 T). Through gating, a tunnel conductance ratio of 35 % for the two magnetization configurations (Fig. 8(d)) has been attained, with a high hysteresis in gate voltage. The conductance ratio can be increased significantly by utilizing thicker barriers. Fig. 8(f) shows the conductance of device based on four-layer CrI_3 tunnel barrier; 400 % conductance was attained at uniform magnetic field of 1.77 T. Each additional layer of CrI_3 functions as a spin filter in series, increasing the conductance ratio among various magnetization structures as the layer count rises [129]. Therefore, this device design could offer a fresh way to investigate non-volatile reconfigurable memory and logic functions [244].

Spin valve/Magnetic tunnel junction (MTJ)

A tunneling magnetoresistance (TMR)/Magnetoresistance (MR) sensor or fundamental magnetic tunnel junction (MTJ)/vertical spin-valve (VSV) comprises of stacked ferromagnets (FM) and insulators, with the efficiency or sensitivity being determined by the condition of the FMs-insulator junctions [245–251]. The junctions are impeded by the peculiar hybridization of non-magnetic (NM) and FM atomic orbitals, which results in spin polarization, and this is referred to as 'spinterfaces'. It was previously stated that ferromagnetic/spacer layer junctions in MTJs/VSVs give rise to TMR/MR and spin polarization, and that the term "spinterface" was developed a time ago to characterize spin polarization induced by the junction between FM material and

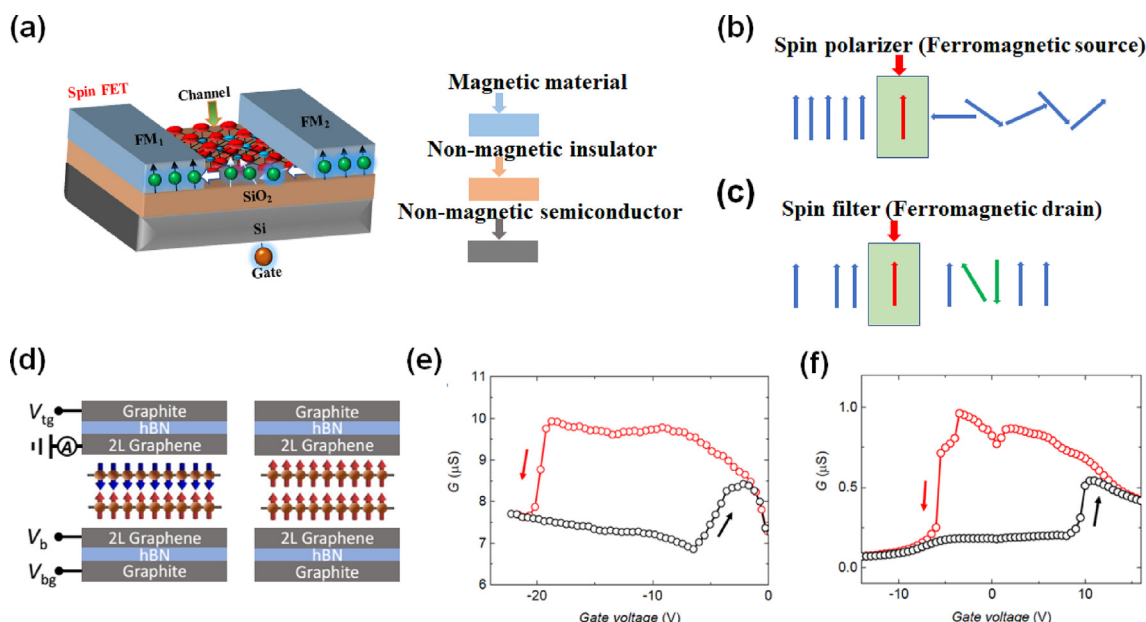


FIG. 8

(a) Schematic diagram of spin-FET. (b) FM source portraying as a spin polarizer. (c) FM drain working as a spin filter. (d) Schematic of spin tunnel FET-based CrI_3 . The arrows represent spin orientation in bilayer- CrI_3 . The left/right sides are related to low/high tunnel conductance states. (e) The tunnel conductance of spin TFET with CrI_3 as spacer layer is repetitively changed by gating as subjected to constant magnetic bias of 0.76 T. The bottom/top gate voltages are equal, and their collective behavior is represented in the horizontal axis. Red and black marks related to measurements as sweeping the gate voltage forward/back, correspondingly. A variation of 35 % is noticed in the tunnel conductance. (f) The conductance of spin TFET is based on four-layers of CrI_3 under a uniform magnetic field 1.77 T. A variation in tunnel conductance is observed about 400 %. This figure is reproduced with the consent of reference [129], copyright, Nature Electronics.

an organic layer [245,252–255]. The spin-valve device uses two FM electrodes and a sandwiched spacer thin layer to recognize spin signals [251,256,257]. The lateral spin valves (LSV) and vertical spin valves (VSV) are two well-known types of spin valves. In a VSV arrangement, the nonmagnetic layer is sandwiched between two FM-layered materials, and the current flows normal to the plane, therefore, preserving the current perpendicular to plane (CPP) configuration as shown in Fig. 9(a). The FM layered materials are put on top of the spacer layer in the lateral arrangement, and the spin current flows laterally, forming current in the plane (CIP) layouts. Fig. 9(b) depicts the CIP setup; the section FM-NM depicts the interface between FM and non-magnetic material, allowing spin to transfer from FM_1 to FM_2 via NM, commonly known as the lateral spin valve (LSV) formation.

The MR of the spin valve is determined based on the positioning of the spins of the two FM electrodes, and it may be precisely controlled by varying the magnetic field. The spin-polarized current is added through the interlayer from one FM electrode (polarizer) to the other FM (detector), showing the low and high resistance states of the device [258–260]. When an in-plane signal is utilized to achieve antiparallel and parallel magnetization states, the resistance value becomes high for antiparallel and low for parallel magnetization alignment among both FM electrodes. The MR/TMR is the identical difference between the resistances obtained in these two designed states, and it is generally expressed as a percentage. The TMR is calculated by using the following relation [261].

$$TMR = \frac{R_{AP} - R_P}{R_P} \quad (1)$$

R_{AP} refers to antiparallel resistance, while R_P refers to parallel magnetization alignment resistance. Here is a preferred and convenient method for estimating the TMR's assessment to the spin polarization of both FM substances for P_1 and P_2 is given as [261].

$$TMR = \frac{2P_1P_2}{1 - 2P_1P_2} \quad (2)$$

Tunneling spin polarization is widely known to be an inherent property of the FM alone, but it is dependent on the electrical and structural properties of spacer layers and their interaction [254,262]. The MR findings for spin up and spin down are examined by the different DOS. If the magnetization orientations for the electrodes are parallel, there will be a large current (low resistance), and the two FM materials will have symmetric DOS for each spin. Asymmetric DOS, alternatively, is produced by the antiparallel arrangement of the magnetization of two FM materials (high resistance) [102]. The summarized data for the various devices that MR is shown in Table 2.

Graphene and chalcogenide semiconductors, which are 2D vdW layered NM materials, intriguing options for tunnel barriers in MTJ devices [263–267]. Spin-valve devices based on graphene were subsequently investigated through experimentation, although the observed MR was less than 5 % [268]. Additionally, it was projected that $\text{Fe}/\text{MoS}_2/\text{Fe}$ devices with a 7-layer MoS_2 spacer layer would have a maximum MR of about 300 % [269]. Experiments devices constructed on various 2D materials, likewise, WSe_2 , WS_2 , BP, MoSe_2 and hBN, as the spacer layer, and typical FM materials as the electrodes, exhibit round 6 % MR [270–274]. The 2D vdW intrinsic ferromagnetic metal Fe_3GeTe_2 (FGT) has been received a great interest due to its metallic trend,

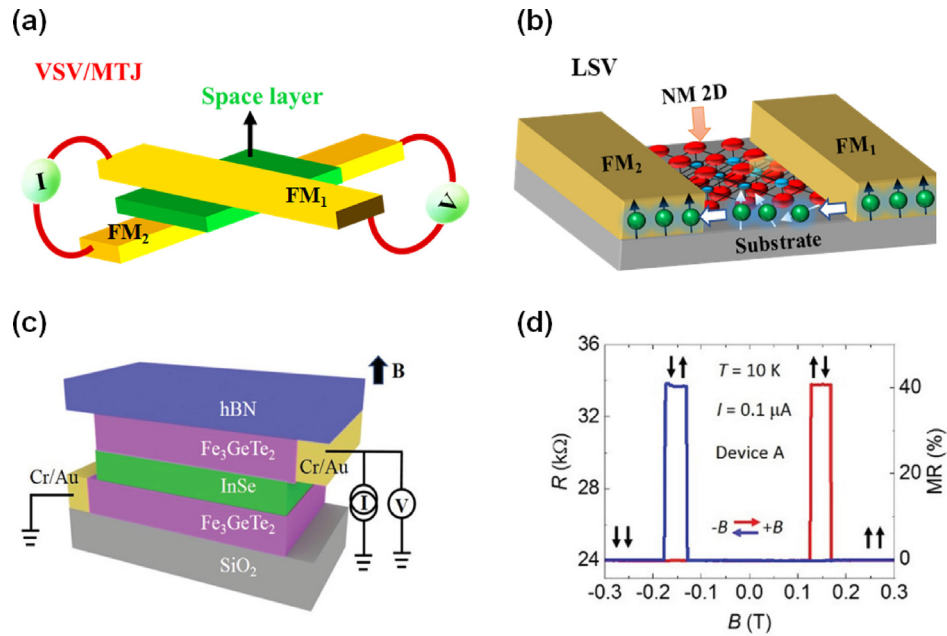


FIG. 9

(a) Schematic arrangement of VSV/MTJ, where the thin layer (NM) is sandwiched among the two ferromagnetic (FM) materials. (b) The lateral arrangement of spin valve (LSV) Here, the FM material is placed on non-magnetic (NM) material through which the spin signal travels laterally. (c) Schematics show the structure and MR measurement frame. (d) Resistance and MR (%) of FGT/InSe/FGT heterostructure with uniform biasing current $0.1\mu\text{A}$ at 10 K. Figure (c) and (d) are imitated with the agreement of the reference [301], copyright, Advanced Materials 2021.

TABLE 2

Summarized data for various VSV/MTJ based on different magnetic materials.

Structure	Year	MR/TMR	Fabrication Method	Reference
NiFe/FLG/NiFe	2014	4.58 %	Exfoliation	[285]
$\text{Fe}_3\text{O}_4/\text{MoS}_2/\text{Fe}_3\text{O}_4$	2015	0.2 %	Sulfurized Mo	[302]
NiFe/ WS_2 /Co	2016	0.47 %	Exfoliation	[424]
NiFe/h-BN/Co	2016	1 %	Exfoliation	[303]
FGT/hBN/FGT	2018	160 %	Exfoliation	[278]
Gr- CrI_3 -Gr	2018	19000 %	Exfoliation	[238]
$\text{VSe}_2/\text{MoS}_2/\text{VSe}_2$	2019	846 %	Theoretical	[248]
$\text{CrTe}_2/\text{Gr}/\text{CrTe}_2$	2020	400 %	Theoretical	[304]
FGT/ MoS_2 /FGT	2020	3.1 %	Exfoliation	[250]
FGT/ WS_2 /FGT	2021	0.45 %	Exfoliation	[305]
CrTe_2 -Gr- CrTe_2	2021	6962 %	Theoretical	[304]
Boron doped graphene				
FGT-InSe-FGT	2021	41 %	Exfoliation	[301]
FGT/ WSe_2 /FGT	2022	164 %	Exfoliation	[297]
FGT-hBN-FGT	2023	65 %	Exfoliation	[306]
FGT-hBN-FGT	2023	480 %	Exfoliation	[299]
$\text{Cr}_{0.85}\text{Te}/\text{MoS}_2/\text{Fe}_3\text{GeTe}_2$	2023	-21 %	$\text{Cr}_{0.85}\text{Te}$ (CVD) $\text{MoS}_2/\text{Fe}_3\text{GeTe}_2$ (Exfoliation)	[307]
1 T-MnSe ₂ /h-BN/1T-MnSe ₂ /h-BN/1T-MnSe ₂	2023	$2.56 \times 10^3\%$	Theoretical	[308]
$\text{Fe}_3\text{GeTe}_2/\text{GaSe}/\text{Fe}_3\text{GeTe}_2$	2023	192 %	Exfoliation	[300]

significant PMA, and greater Curie temperature (T_C) up to 220 K [70,275–277]. Tunneling devices based on structure FGT-hBN-FGT were investigated previously, with a high MR of 160 % obtained [278]. The spin valve composed of FGT/ MoS_2 /FGT was devised by Ziao Wang *et al.* They reported MR of 3.1 % at 10 K, which is around 8 times more than that of the known spin valves made of MoS_2 sandwiched by traditional ferromagnetic electrodes [250]. In FGT- MoS_2 -FGT structure based devices, the semiconductor MoS_2 has also been introduced as a spacer layer,

however, it works as a conductive layer instead of a tunneling barrier [250]. Various VSVs and MTJs were explored earlier based on different spacer layer of oxides [279], 2D materials [264,268,280–287], TMDCs materials [288], and their heterostructures as well [241,289,290]. In addition, the FGT-InSe-FGT structure based device is expected theoretically to be a flawless spin-filter with a massive TMR ratio about 700 % [291]. We reviewed here, the 2D FGT/InSe/FGT VSV device, whereas the schematic design of the FGT-InSe-FGT device is

shown in Fig. 9(c). The junction region is copped with hBN to prevent contamination and degradation in the air, and the layered material InSe is placed among two FM metallic FGT-based electrodes. To regulate the magnetization direction of the two FGTs during the magneto-transport experiments, a magnetic field (**B**) that was directed normal to the plane is being used. The FGT's coercivity is influenced by both the layer thickness and its shape [277,292]. The FGT flakes of various thicknesses were thus used to fabricate the heterostructure in order to attain drastically varied coercivities among the top/bottom FGT electrodes. The I-V curve of device at low temperatures (10 K) was non-linear, indicating that a tunneling phenomenon may have been developed between spacer layer InSe and the FGT FM electrodes. On the other hand, MTJs with a thin spacer layer can have their MR affected by pinholes. Since, few electrons travel from the top/bottom electrode instead of tunneling through the spacer layer when pinholes are present, the MR is reduced [293,294]. Pinholes can be formed as a result of the spacer layer's surface roughness, whose thickness can be decreased in some places to increase electron transmission [295]. As a result of the creation of a pinhole-free tunnel barrier at the $\text{Fe}_3\text{GeTe}_2/\text{InSe}$ junction and pinholes in the InSe tunnel barrier, correspondingly, two unique transport behaviors; tunneling and metallic are observed. The devices achieved a high MR of up to 41 % with applied currents of less than $0.1 \mu\text{A}$ at 10 K as shown in Fig. 9(d). The device's MR is extra robust to bias current than the metallic one, allowing it to function at lower bias currents. Both device's MR ratios reduced with rising of temperature, and the improvement of spin polarization with temperature may be fairly matched by Bloch's equation [88]. The presence of pinholes in the InSe layer is responsible for the lower MR in the other metallic device. These findings demonstrate how pinhole-free spacer layer-based all-2D vdW heterojunction structures may greatly improve the MR and serve as an attractive platform for the integration of spin logic and nonvolatile memory [266,296]. Wen-kai Zhu *et al.* reported large TMR up to 85 % at room temperature ($T = 300 \text{ K}$) in vdW MTJs based on FGT/WSe₂/FGT. Additionally, the TMR is enhanced up to 164 % at $T = 10 \text{ K}$ [297]. One more thing, based on two few-layer exfoliated same FGT flakes, wang *et al.* constructed a facile vertical homojunction. By supplying a negative or positive pulse current, which may also diminish the coercivity through the SOT of the FGT itself in accumulation to the Joule heat, it is possible to accomplish magnetization reversing under a particular range of external magnetic fields. Moreover, the SOT in the homojunction is what causes the asymmetrical switching current. The magnetization inversion may be seen with a weaker external magnetic field as the temperature rises [298]. Furthermore, recently, for the device with a 2° twist angle, the tunneling TMR of the FGT-hBN-FGT spin valves is observed to be as large as 480 % and steadily drops to 100 % and decreases as the misalignment angle grows. These novel twist-angle dependent spin-valve activities give low-dimensional spintronic applications a power management grip [299]. While reducing the lattice-matching necessities from the epitaxial growth and facilitating high-quality incorporation of distinct materials with atomically sharp interfaces, using atomically thin vdW materials in MTJs is anticipated to improve their figure of merit, the tun-

neling TMR. Recently, in vdW $\text{Fe}_3\text{GeTe}_2/\text{GaSe}/\text{Fe}_3\text{GeTe}_2$ MTJ, TMR up to 192 % at 10 K is observed [300]. This proved that the applied bias or semiconductor thickness may be used to modify both the magnitude and the sign of the TMR, highly allowing spin-polarized carriers in vdW semiconductors to be modulated.

Spin-orbit torque (SOT)

For controlling magnetization in non-volatile magnetic memory systems, spin-orbit torques (SOTs) have gained a great deal of attention. The efficient modulation of magnetization by electric current using spin-orbit torques (SOTs) has applications in non-volatile memory and logic mechanisms. In a non-magnetic/ferromagnetic heterostructure, the applied electrical current, in the non-magnetic layer generates a spin current, which is injected into adjacent ferromagnetic layer [309,310]. The devices with high SOT efficiency often use an NM material with high SOC in close proximity to an FM material. The SOT is a novel method for modifying magnetization in FM materials that requires introducing an in-plane electric current all the way through a high SOC material [311]. In an SOT device, the current passing through a bilayer configuration made of FM and heavy metal can induce significant changes in the produced magnetization. The intrinsic spin Hall effect (SHE) and the superiority of the junction both have a role in the switching of the magnetization in traditional SOT bilayer structures [312,313]. SOT has a lot of potential for energy efficiency, which makes it desirable. A dimensionless integer that expresses the amount of angular momentum transferred per electron can be used to represent the efficiency of creating spin current and transferring angular momentum. The SOT-based devices frequently have separate write and read channels, such as SOT memory. Neuromorphic circuits [314,315], MRAM [98,316,317] and nano-oscillators [318,319] are just potential uses for SOT.

The frameworks for comprehending SOTs are provided by two basic models. The first concept is that SOTs form in a bi-layer structure comprising a NM layer and an FM layer from spin currents formed in the NM layer and linked to the FM layer at the junction due to the SHE [320,321]. The SHE produces a spin current that travels to the junction, producing spin torque comparable to that generated by a perpendicular spin-polarized current as presented in Fig. 10(a). The second is that SOTs can occur because of the interaction of SOC and magnetic switching at a junction among a magnetic layer and an NM layer, as postulated by Manchon and Zhang [322] based on the Rashba-Edelstein effect [323]. The current interacts with the SOC in a 2D electron gas (2DEG) with magnetic exchange and SOC to generate a spin accumulation that applies a torque on magnet as illustrated in Fig. 10(b). A SOT device made of single-layer TMDCs (MX_2 , $\text{M} = (\text{W}, \text{Mo})$, $\text{X} = \text{Se}, \text{S}$ and FM CoFeB bilayer was described by Shao *et al.* [324]. Recently, Ostwal *et al.* presented a device using 2D magnetic materials, $\text{Cr}_2\text{Ge}_2\text{Te}_6/\text{Ta}$ based device [325], the magnetization in CGT is guided by the collective in-plane magnetic field and charge current passing through tantalum (Ta). The benefit of 2D magnetic materials for the discovery of low-utilizing and highly condensed SOT devices was evident from the switching current densities of CGT/Ta SOT devices being significantly smaller than those of other 3D FM based devices, such as CoFeB. Categorically, an SOT device is more auspicious if both opera-

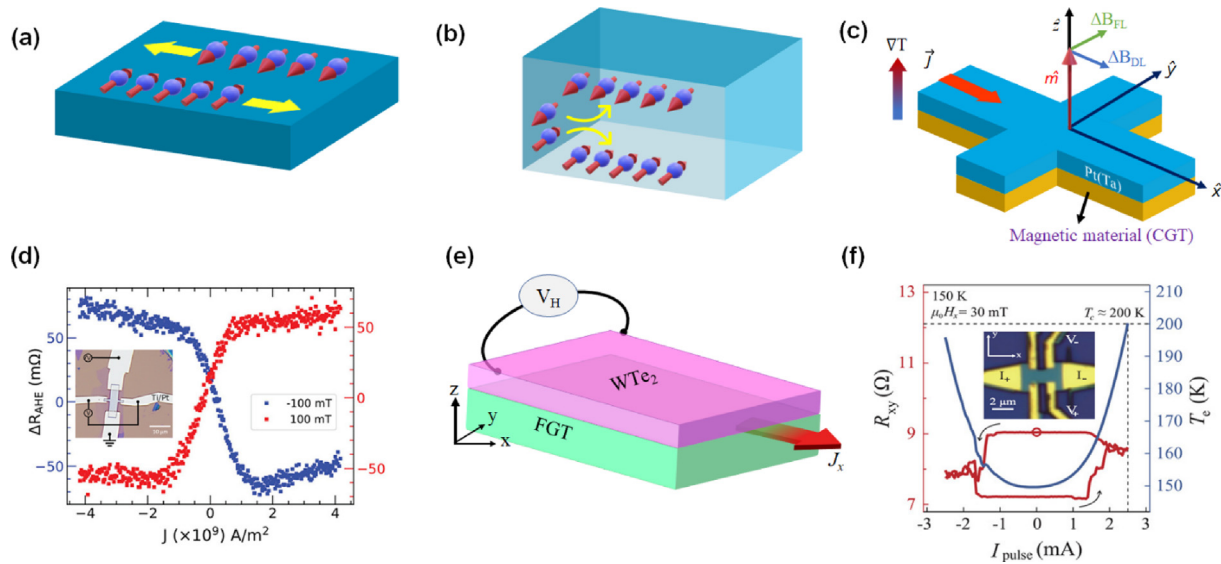


FIG. 10

Schematic diagram of (a) SHE. (b) Rashba Edelstein effect (c) Schematic diagram to observe SOT, additionally the investigation of thermomagnetic phenomena, representation of alignment of effective field. (d) SOT switching of the magnetization in CGT/Ta in in-plane magnetic field $B_x = \pm 100$ mT, inset: displays the optical image of the device which shows the thin layer of CGT and Ta Hall bars. Figures (c) & (d) are imitated by the agreement of the reference [340], Copyright Nano letter, ACS, 2020. (e) Schematic diagram of the structure of FGT/WTe₂ magnetic characteristics measurement approach, the charge current (J_x) is applied to WTe₂ to produce a spin current that is inserted into FGT. (f) Current-stimulated magnetization switching in FGT/WTe₂. Hall resistance R_{xy} (red line) was examined after applying current in the existence of the in-plane magnetic field (30 mT) parallel to charge current at temperature 150 K. The blue line shows electron temperature (T_e) during current sweeping is shown as a function of I_{pulse} . These figures (e) & (f) are reused with the approval of the reference [353], Advanced Materials (2022).

tional layers can be constructed from 2D materials. TMDCs are interesting contenders for creating these torques since they owe both extremely greater charge-to-spin transformation efficiencies and magnetization modulation symmetries and orientations [326,327]. Furthermore, TMDCs have a variety of appealing features, including as strong SOC, good crystalline feature, and various crystalline regularities [328,329]. Materials having extremely greater charge-to-spin transition efficiencies, such as WTe₂ and TaTe₂, are frequently seen as promising possibilities for high SOT proficiencies [327,330]. Substantial charge-to-spin transformation efficiencies, however, are not a surety of large spin-orbit transfer (SOT) efficiencies because SOTs are frequently a developing singularity that depends on proximity effects, wavefunction overlap, and interface spin transparency (spin mixing conductance), among other factors. For greater SOC materials, consisting of heavy metal, topological insulators (TIs) [331,332], and newly, topological semimetals (TSMs) [224,333–335], have been investigated to improve their spin Hall angle $\theta_{\text{SH}} = J_s/J_c$, an observation of their efficiency at transforming charge current density J_c to spin current density J_s . Additionally, interface engineering has been researched to optimize the spin transparency (T_{int}), across the junction between both layers of FM materials [336,337]. The main issue for efficient SOT spintronic devices is to enhance the SOT efficiency [96] as given in equation (3)

$$\xi = \theta_{\text{SH}} \cdot T_{\text{int}} \quad (3)$$

Yet, FM and topological vdW constituents have been utilized as one of the fundamental layers, either one for spin-current production or FM layer such as topological insulator (TI) Bi₂Se₃ with

FM layer of CoFeB [332] or metallic magnetic material Fe₃GeTe₂ (FGT) with a linkage with heavy metal platinum (Pt) film [338,339].

Here, we looked at observations of SOT effects and thermoelectric brought through current in devices that incorporate semiconducting Cr₂Ge₂Te₆ (CGT) magnetic material with Pt and Ta metal junction [340]. Here multilayer CGT/Ta and CGT/Pt samples were utilized, the magnetic alignment of magnet is investigated by Anomalous Hall effect (AHE) through heavy metal and observation of thermomagnetic effects was explored in these experiments [192,341,342]. This approach offers a way to observe SOT in CGT/Pt structure, the torque efficiency is very similar to those found by other structures that are based on combination of Pt with conventional 3D magnetic materials. Fig. 10(c) shows the schematic diagram of the general device, which shows the direction of the fields because of current-stimulated damping-like torque (ΔB_{DL}) and field-like torque (ΔB_{FL}) working on magnetization vector $\vec{m}$. Fig. 10(d) represents the SOT switching of magnetization, this illustrates the observed Hall resistance of the CGT (8.9)/Ta (6 nm) sample as sweeping a dc current utilized. The inset: displays the optical image of the structure of the device with Hall bars. The polarization of the current-stimulated switching is reversed by inverting the alignment of the in-plane field. These investigations are anticipated for switching because of SOT [196,198,343] and validate the findings reported in ref. [325]. In comparison, SOT applied by heavy metal on the thin film of 3D PMA materials [198,343–346], the critical current densities for switching CGT are lower by one order. The thermoelectric signals and Joule heating in CGT-based samples [325,341] can generate back-

grounds in observations of SOTs [347–350] that must be taken into consideration to examine the torques perfectly. Furthermore, thermal phenomena were also explored in this report [340], when compared to magnetoresistance signals, the intensity of thermoelectric signals in these structures may be rather strong, with second-harmonic Hall tests being significantly influenced by both the conventional Nernst effect and the longitudinal spin Seebeck effect. The strong spin Seebeck signals point to CGT/heavy-metal materials as potentially fruitful research platforms for studying the behavior and transport of magnons in vdW magnetic materials. These findings show the potential of adding semiconducting 2D magnets into spin-orbitronic and magneto-thermal systems.

A spin-current with a high SOT efficiency (ξ) and electrical conductivity (σ), as well as an effective spin insertion through a transparent junction, are necessary for energy-efficient spintronic devices. Here, single crystals of the vdW-heterostructure with an atomically precise interface vdW semimetal WTe_2 and vdW FM Fe_3GeTe_2 are employed to meet the specifications. However, we reviewed here SOT that initiated from the current driven through WTe_2 which can significantly switch the FGT magnetization of the device in the existence of H_x . Besides the atomically flat interface and large SOC, WTe_2 can also provide an out-of-plane polarized spin current to achieve field-free switching, which is also an advantage compared to the traditional heavy metal. The schematic diagram of the structure based FGT/ WTe_2 is shown in Fig. 10(e). This represents the SHE in WTe_2 creating a pure spin current, which is inserted into FGT layer and applied a SOT on the magnetization of FGT. A current I_{pulse} is applied and R_{xy} was examined as shown in Fig. 10(f). The switching of R_{xy} occurred at $I_{\text{pulse}} = \pm 15$ mA. Commonly, the charge-to-spin conversion efficiency is computed by the SOT efficiency equation (4) [339,351].

$\xi = \frac{4e}{\pi\hbar} M_{\text{FGT}} \cdot t_{\text{FGT}} \frac{H_z^{\text{coer}}}{J_{\text{sw}}} \quad (4)$. where, M_{FGT} is the magnetization saturation of FGT, t_{FGT} represents the thickness of FGT. H_z^{coer} is the coercive field along easy-axis, and J_{sw} is the switching current density of the device. These outcomes show the value of $\sigma = 2.25 \times 10^5 \Omega^{-1} \text{m}^{-1}$ and $\xi = 4.6$ for this structure. Furthermore, the substantial reduction in switching current density were attained about $3.9 \times 10^6 \text{Acm}^{-2}$ at 150 K which is much lower in magnitude than other conventional ferromagnetic/heavy metal films. To properly transition the magnetization from one state to another electrically when creating spintronic nanodevices like magnetic memory, 2D vdW magnets are still needed. Xiao Wang *et al.* developed a bilayer framework of FGT/Pt in which the SOTs initiating from the current flowing in the Pt layer may efficiently alter the magnetization of few-layered FGT. Using harmonic evaluations, the effective magnetic fields related to the SOTs are further quantitatively described [338]. Additionally, at ambient temperature, spin-orbit torques (SOTs) may alter the perpendicular magnetization of the 2D vdW FM FGT with a relatively minimal current density of $1.3 \times 10^7 \text{A/cm}^2$. Additionally, harmonic measurements are employed to objectively determine the high SOT efficiency of $\xi \sim 0.22$, which is greater than that of Pt-based heavy metal/conventional FM devices [352]. These results show that developing vdW-type topological materials and magnets offers a potential path to SOT-based spintronic-energy-efficient magnetization manipulation [353].

Magnetic random-access memory (MRAM)

MRAM is a form of non-volatile memory that stores data in magnetic domains [98,354–358]. The non-volatile MRAMs such as spin-transfer torque (STT) MRAM and next-invention SOT MRAM are developing as critical to facilitating low-power advancements that are predicted to proliferate across vast industries ranging from data storage to the Internet of Things (IoT). Simultaneously, the invention and implementation of devices utilizing 2D vdW heterostructures enable the production of ultracompact multilayer composites with previously unintelligible material-engineering capabilities. Here, we give a general summary of the advancements and difficulties now facing MRAM, as well as the potential benefits that might come from combining 2D material technology. We identify the essential characteristics of atomically smooth surfaces, decreased material mixing, crystal regularities, and proximity effects as the leading factors behind potential game-changing advancements for MRAM at advanced technological stages. However, it is anticipated that the MRAM industry will increase greatly in the near future by tackling power and transfer of data (memory-wall limitation) concerns in a number of embedded applications, as well as biological systems and in-memory computing [359,360]. To expand, MRAM technology must overcome many hurdles that necessitate material and device integration advances, with 2D materials potentially playing a crucial role. Certainly, since the innovation of graphene, which has been observed by ever-increasing 2D materials have emerged as possible facilitators for a family of related materials [361]. Ultracompact device designs or fundamentally novel concepts are possibilities for spinning data processing [362,363]. Scientific developments in 2D-based spintronic devices, including recent advancements in large-scale co-integration of 2D with traditional microelectronics materials [364,365], have paved the way for the development of new MRAM technology. We present an outline of the most contemporary discoveries, outline the problems and possibilities presented by 2D materials for MRAM, and provide a perspective of how advances in 2D materials research might transform into a game-changing solution for upcoming productions of spintronic-memory technology.

STT and SOT-based M-RAM

MTJs, which comprise of two FM layers separated by a thin insulator layer, are the foundation of non-volatile MRAM technology. The free layer (FL), one of the FM layers, is used to store data, while the reference layer (RL), another layer, offers a reliable reference for accessing the FL data. STT [366] and SOT [310] are the two basic methods for controlling FL magnetization that rely on electrically produced spin currents. In a two-terminal architecture, STT is seen when the MTJ receives an injection of vertical current shown in Fig. 11(a). Due to the transfer of angular momentum, the current tunneling through the spacer layer is spin-polarized, which causes the FL magnetization to reverse the sufficiently high currents. The resistance of the device is determined by the relative magnetic alignment of the FL and RL layers, which is tracked by the TMR. SOT-MRAM is a three-terminal device with an extra conducting channel beside the FL shown in Fig. 11(b). Due to the spin hall effect (SHE) and/or Rashba-Edelstein effects, a charge current running along this

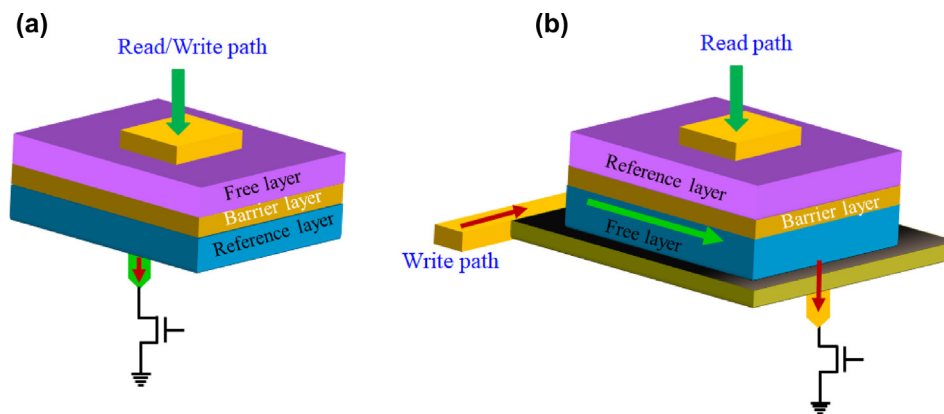


FIG. 11

(a) Schematic illustration of STT-MRAM. (b) SOT-MRAM, which shows the read/write paths.

channel causes the addition of a spin current into the FL [198,310,367,368]. The SOT can alter FL magnetization in such devices without allowing a current to penetrate the MTJ tunnel barrier. The decoupling of the write/read current channels minimizes write mistakes for the duration of the reading process and lowers the danger of voltage breakdown at the cost of adding another terminal. It also allows for more variable cell layouts by enabling the spin torque direction to be controlled freely of the anisotropy direction in the stack [369]. Material advancements have made significant progress in spin-torque memory devices. Several excellent material patterns have been established, with CoFeB/MgO playing a key role for several years. In recent years, a wide range of innovative developing 2D materials and heterostructures have demonstrated the capability of addressing the aforementioned concerns and obstacles. Furthermore, because they are atomically thin, their characteristics might be modified by external electric fields or proximity effects [363].

Magneto-Memristive switching in 2D material for neuromorphic computing

A memristive device whose resistance may be hysterically controlled by an electric field or current is being sought after for its theoretical merits as well as for prospective uses in phase-change memory and neuromorphic computing. When the underlying material shows spin order, the resistive states can be coupled, further enhancing the electrical control over the collective phases. This report indicates the discovery of rapid, memristive tunneling current switching in nanoscale junctions of ultrathin CrI₃, a natural layer antiferromagnet. Even in multi-layer samples, the coupling to spin order allows for the magnetic field-tuning of resistance hysteresis and the electric-field switching of magnetization. In 2D magnetic materials as CrI₃, it was investigated earlier, high electric fields cause the tunneling current in graphene Gr-CrI₃-Gr structure to switch robustly and hysterically, which is a common characteristic for a locally active memristor [370]. The magnetic field can be utilized to modulate the hysteresis of the resistive transition, although the electric field can also be utilized to tune the magnetization across the layers of magnetic materials. The approach is not only innovative to explore 2D CrI₃ but may significantly generalize to antiferromag-

nets at higher or room temperatures [371]. Fig. 12(a) shows the schematics of the device that consisted of thin magnetic material which is sandwiched between a top and bottom graphene (Gr) and the whole structure is encapsulated with h-BN. Using this approach the CrI₃ was sandwiched with graphene previously [371]. The junction area was selected very small to attain robust characteristics through this device. Fig. 12(b) demonstrates a key feature of this experiment which was achieved by using 16-layer of CrI₃ based at 1.4 K. The voltage (0.03 mV s^{-1}) must be ramped up gradually until it reaches the high, critical value, V_C^H , or else the tunnel current will suddenly increase by more than a factor of two. A subsequent voltage rise reveals a new state with large conductivity. The current quickly drops at a lower significance threshold, V_C^L , as the voltage ramps down from this stage, but it then resumes its initial, lower conductance state. These critical voltages can be regarded as intrinsic characteristics of the device because they are not affected with ramping speed by more than five orders of magnitude [370,372]. This work illustrates a nanoscale memristor device with instantaneous configurable resistance switching by magnetic fields and tunable magnetization by electric fields. This induced mechanism is entirely generic, even if functioning is currently limited to low temperatures, and it may be applied to a variety of materials. For functioning at high temperatures, specifically, synthetic antiferromagnets [371] with layer-dependent magnetic ordering like CrI₃ but greater T_c may be used as a replacement. In addition to improve the functionalities of 2D magnets, the proposed device design also offers a fresh perspective for further research into neuromorphic computing [373].

Material engineering

Due to the suitability of 2D magnetic materials for device construction, 2D flake's magnetic characteristics can be tuned externally using factors like electrostatic gating and fields [9,43,374,375], strain engineering [376,377] and pressure [378–380]. Furthermore, extrinsic perturbations like the electric field, doping, chemical functionalization, and stacking engineering may effectively alter the magnetism in these ultrathin magnets. The long-range FM order of the CrI₃ monolayer produced in a recent experiment has garnered a lot of interest. However, its practical applicability is severely constrained by the extremely

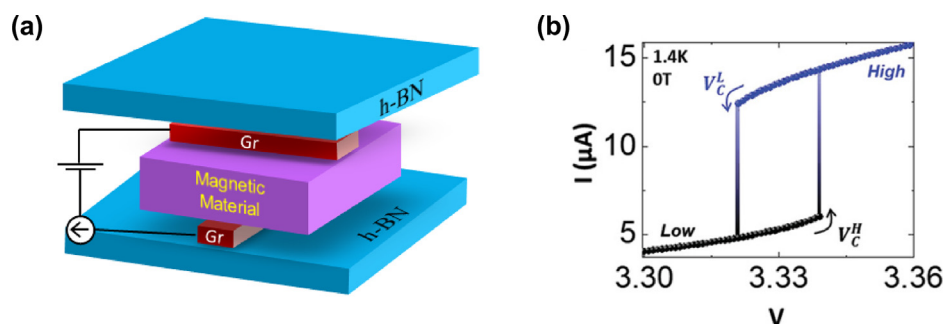


FIG. 12

Memristive switching in magnetic material-based tunnel junction. (a) Schematic representation of the device based on memristive switching through magnetic material, CrI_3 is sandwiched between top and bottom graphene (Gr). (b) Demonstration of rapid and hysteretic switching of tunneling current across multilayer CrI_3 at 1.4 K. This figure is utilized here, by the permission of the reference [371], copyright, *Advanced Materials* 2020.

low Curie temperature T_c (~ 45 K). Here in the study (exploring magnetic stability and valley splitting on CrI_3/SiC van der Waals heterostructure), we reviewed here, how to build a vdW heterostructure for CrI_3 with a monolayer of SiC to see how magnetic characteristics may be improved [381]. According to the precise calculations, both the proximity exchange effect and the super-exchange interaction may increase the FM coupling of the CrI_3 layer in a heterostructure with T_c up to 62.3 K. A large magnetic exchange splitting of the valley is also present in the inherently nonmagnetic SiC layer, which is likewise magnetized. More significantly, it has been determined that the magnetic exchange energy progressively grows as the interlayer gap narrows, resulting in a T_c of up to 128.8 K, and that the external electric field improves the heterostructure FM coupling as well, resulting in a T_c of up to 93.14 K. These developments offer various ways to enhance the magnetic stability of two-dimensional magnetic materials and introducing magnetism into other non-magnetic materials.

However, for nanoscale spintronics ferromagnetic 2D semiconductors are crucial and have great demand, but we highlighted here intrinsic ferromagnetism in an emerging family of single-layer 2D semiconductors called CrOX (CrOCl and CrOBr monolayers). These materials exhibit strong ferromagnetic ordering, significant spin polarization, and high Curie temperatures. From their bulk antiferromagnets, these 2D crystals offer excellent dynamical and thermal permanence as well as simple experimental fabrication. The Curie temperature of 2D CrOCl is 160 K, exceeding the previous high (155 K) set by the most extensively researched dilute magnetic GaMnAs materials. This temperature might be raised much higher with proper strain [382]. Based on density functional theory, a 2D chromium oxide fluoride CrOF film was investigated. According to the study, CrOF film is a naturally ferromagnetic semiconductor. The Curie's temperature is 150 K in this case. Particularly during hole doping, the Curie temperature rises to 410 K and the CrOF sheet turns into a half-metal. The 2D CrOF sheet might be a good choice for next-generation spintronic devices due to its diverse electrical and magnetic characteristics [383]. This research makes 2D CrOX monolayers an excellent foundation for spintronics in the future.

To realize sophisticated spintronic and magneto-electronic capabilities, two-dimensional (2D) materials must have magnetic phase control and room temperature magnetic stability. In

recent reports (Bi-stimuli assisted engineering and control of magnetic phase in monolayer CrOCl) investigating the effects of strain and electric field on the 2D material, CrOCl, have shown that uniaxial strain enables the possibility of room-temperature ferromagnetism in the layer and also detected the development of an anisotropic ferromagnetic to antiferromagnetic phase change in the system. In addition to this, the correlation of strain and electric field causes a notable increase in the Curie temperature ($T_c = 450$ K) of CrOCl. These simulation-based predictions highlight the possibility of multi-stimuli magnetic phase control, which might be extremely important for the implementation of magneto-mechanical sensors [384].

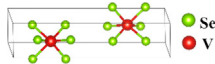
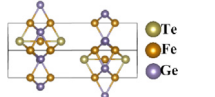
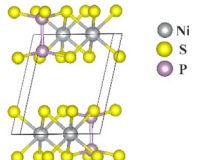
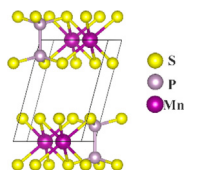
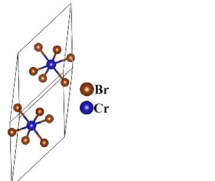
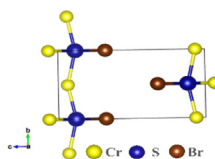
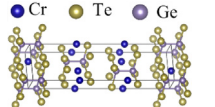
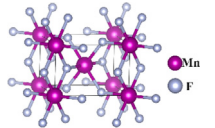
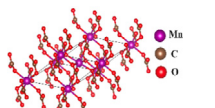
Since, the interlayer exchange coupling (IEC), as demonstrated by Yi Cao *et al.*, enables the Curie temperature and PMA of vdW Fe_3GeTe_2 to be modified [385]. With Pt and Ru spacer layers, they showed strong ferromagnetic and antiferromagnetic IECs among sputtered Co and vdW Fe_3GeTe_2 . It was found that ferromagnetic IEC maintains PMA, which may demonstrate 100 % out-of-plane remanent magnetism above room temperature, despite raising the Curie temperature of Fe_3GeTe_2 . These 3D/2D interlayered linked hybrid structures may prove to be the best choice for real-world vdW spintronic applications [385].

The charge transfer phenomenon allows for electrical tuning of magnetic materials. Charge transfer has significance at heterostructure interfaces because it may influence electrical structures and eventually the physical characteristics of the materials. However, once the contact has been constructed, charge transfer is challenging to manage outside. An ideal platform for the electrical regulation of interfacial charge transport was offered by the recently found vdW's ferromagnets with atomically sharp surfaces. To selectively alter the switching fields of the top or bottom Fe_3GeTe_2 electrodes, Ziao Wang *et al.* conducted magnetoresistance studies demonstrating electrically adjustable charge transfer in FGT/CGT/FGT all-magnetic van der Waals heterostructures [386]. First-principles calculations demonstrate the directed charge transfer from metallic FGT to semiconducting CGT, which noticeably alters the magnetic anisotropy energy of FGT and causes drastically reduced coercivity. This study's electrically selective manipulation of magnetism may encourage the creation of spintronic devices using vdW's magnets.

Furthermore, we added details of various ferromagnetic (FM), and antiferromagnetic (AFM) materials, with their crystal struc-

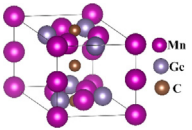
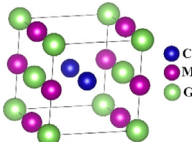
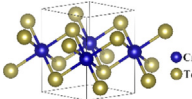
TABLE 3

Summary of various magnetic materials with several characteristics.

Materials	Curie/ Néel Temperature T_C , T_N (K)	Magnetic order & Moment (μ_B)	Bandgap (eV)	Preparation & Crystal System	Crystal Structure illustration	Reference
VSe ₂	300 K	FM 0.731 μ_B	Metallic 0.0 eV	MBE, Mechanical Exfoliation, Trigonal		[236,388]
Fe ₃ GeTe ₂	270 K	FM ⊥ 2.42 μ_B	Metallic 0.0 eV	CVD, Mechanical Exfoliation Hexagonal		[192,389–391]
NiPS ₃	~100–154 K	AFM	1.6 eV	CVT Zigzag and tilted. Monoclinic		[392,393]
MnPS ₃	T_N = 78 K	AFM 4.46 μ_B	2.6 eV	Néel, Isotropic Monoclinic CVT		[394–397 20,398]
FePS ₃	115–118 K	AFM	1.6 eV			
CrBr ₃	20–37 K	FM ⊥ 2.992 μ_B	1.68–2.1 eV	CVD & Exfoliation Trigonal		[399,400]
CrSBr	168 K	FM 0.11 μ_B	1.66 eV	Mechanical Exfoliation Orthorhombic		[401]
Cr ₂ Ge ₂ Te ₆	61 K	FM 0.214 μ_B	0.95 eV	CVD/ Mechanical Exfoliation Trigonal		[402–404]
MnF ₂	Néel Temperature T_N = 67 K	AFM 2.5 μ_B	3.28 eV	MBE Tetragonal		[405,406]
MnCO ₃	300 K	AFM	3.25 eV	Magnetron Sputtering Rhombohedral		[407,408]

(continued on next page)

TABLE 3 (CONTINUED)

Materials	Curie/ Néel Temperature T_C , T_N (K)	Magnetic order & Moment (μ_B)	Bandgap (eV)	Preparation & Crystal System	Crystal Structure illustration	Reference
$Mn_5Ge_3C_{0.8}$	430–450 K	FM $\perp$ 1.19 μ_B	0.5 eV	MBE Hexagonal		[409]
Co_2MnGa	1015 K	FM $\perp$ 4.103 μ_B	0.15 eV	Magnetron Sputtering/MBE Cubic		[189,410]
$CrTe_2$	310 K	FM $\parallel$ $\sim 3 \mu_B/\text{atom}$	Metallic 0.0 eV	CVD/MBE Exfoliation		[411–413 414,415]
CrS_2	~ 1000 K	$\sim 2.09 \mu_B/\text{atom}$	0.0 eV	Trigonal		

ture, Néel Temperature (T_N) and Curie temperature (T_C) as shown in Table 3. The sign $\parallel$ referred to in-plane and $\perp$ shows the out-of-plane alignment of the magnetic field for the magnetic materials. We utilized VESTA software [104,105] to draw different crystal structures of magnetic material and available structure files obtained from the materials project [107,108] and crystallography open database [109,111,112,387].

Challenges and perspectives

This study encompasses the most recent experimental developments in magnetic 2D vdWs materials, especially a massive focus on possible spintronic device applications. In this review, we outlined the recent developments in synthesis, characterization, and spintronic applications of 2D magnetic materials. Despite tremendous progress in this area, several elusions are still existed. Numerous issues need to be resolved, contrary to most 2D magnetic systems, the majority of vdW materials deteriorate in air and have magnetic ordering temperatures below room temperature; they are also vulnerable to oxidation such as $CrGeTe_3$ [50,416] and Fe_3GeTe_2 [417]. It is indispensable to find air-stable, vdW materials that are magnetically ordered at ambient temperatures, especially magnetic insulators, and semiconductors, for practical applications employing magnetic vdW heterostructures. This might necessitate the development of brand-new vdW magnets with greater anisotropies and stronger exchange interactions in the monolayer limit. A promising beginning to solving this problem is the recent finding of itinerant ferromagnetism up to 270 K in ultrathin flakes of Fe_5GeTe_2 [52] and at normal temperature in monolayer $MnSe_x$ [22] and VSe_2 [21]. Encapsulation in insulating h-BN can also enhance the performance of 2D magnetic materials-based devices. However, a low nucleation rate results in the development of huge individual flakes of excellent crystalline quality that cover only a tiny portion of the substrate and have randomly oriented domains and poor layer control. The incorporation of less stable

metal and chalcogen precursors in TMDCs formation lowers the growth temperature (300–560 °C) while increasing nucleation rate [135,418]. The precursor control growth might substantially lower the fabrication's thermal cost, making it a potential option for direct growth in back-end-of-line (BEOL) compatible wafers. TMDCs monolayers are produced on a wafer size using this approach, but they retain polycrystalline, reducing their electrical characteristics. However, discovering new types of 2D magnetic materials with improved stability and performing batch manufacturing remain difficult tasks. The controlled synthesis of novel materials Cr_5Te_8 , $CrTe_2$, CrS_2 , together with its exceptional and air-stable FM characteristic, offers up new avenues for research into 2D magnetism and spintronic device applications. Although several vdW materials are theoretically anticipated to be magnetic in the 3D and 2D limit, only a small number have been confirmed empirically so far. Recently, evasion of detail study has been found on the magnetic characteristics of 2D thin layers and how they compare to bulk form. Several 2D magnetic materials must be synthesized, and it needs to be done at a scale and with good reproducibility and unique architecture having Curie temperatures above room temperature.

So, the requirement for atomic-level accuracy in managing the thickness of oxide tunnel barriers constitutes a significant problem. The spintronics community is working hard to develop and manage ultrathin MgO and TiO_2 tunnel barriers of uniform thickness on an extensive scale [419,420]. One of the primary advantages is the inherent ability of 2D tunnel barriers to be designed layer by layer with atomic accuracy [421–423]. The decrease in MR/TMR is caused by flaws in the tunnel barriers of MgO , TiO_2 , Al_2O_3 , and other materials, as well as lattice mismatch between the tunnel barrier and FM. On the other hand, MJTs with a thin spacer layer can have their reduced MR by pinholes. Nevertheless, the 2D materials with excellent stability suppressed diffusion, and well-defined interfaces which may encourage the MTJ stacks closer to the ideal one and allow for downscaling. Moreover, the MRAM is promoting major develop-

ing markets such as automobiles, wearables, biosensors, the (IoT), and different cache and buffer memory applications. Its growth over the past several years has been outstanding, and it is obvious that it will soon grab a key.

Further, the developments and manufacturability using commercial CMOS-compatible techniques are primary steps for future advancements. Therefore, it seems sensible to think that the broad family of 2D materials may serve as a road map for future developments in non-volatile spin-torque memory technologies.

Conclusion

Conclusively, the most existing scientific advancements in magnetic 2D vdW materials are analyzed in this research with attention to possible spintronic device applications. In addition to extending the synergy between spintronics and 2D materials, as this article has shown, the tremendous advancement and significant accomplishments of 2D-vdW magnets also bring numerous opportunities and problems for the design and manufacturing of 2D spintronic materials and devices. The field of magnetic vdW materials made significant strides in a short period, expanding to encompass a variety of magnetic phases that may be manipulated by temperature and other external degrees of freedom as well as flexible heterostructure engineering. As a result, achieving room-temperature long-range spin ordering in 2D semiconductors is still difficult, and this might be a key area of research in the future. In contrast, even though numerous spintronic vdW heterostructures based on 2D materials have been postulated in both theory and experiments, there is still a significant amount of room for the various stacking and combining of such ultrathin layers, providing a remarkable opportunity to investigate the physical mechanism and device applications in this rapidly expanding field. Consequently, new theoretical and experimental databases for spintronic heterostructures should be constructed for further investigation. However, research on large defects, such as domains, boundaries, and dislocations in 2D magnetic materials is still in its early stages and is challenging to prepare and comprehend the low-dimensional devices for spintronic applications. This report confirms the discovery of rapid, memristive tunneling current switching in nanoscale junctions of ultrathin CrI₃, a natural layered antiferromagnet. For execution at high temperatures, in particular, synthetic antiferromagnets [371] with layer-dependent magnetic ordering like CrI₃ but greater T_c may be used as a replacement. In addition to improving the functioning of 2D magnets, the proposed device design also offers a fresh perspective for further research on neuromorphic computing [373] and robust memory. Overall, 2D magnetic materials show diversity for next-generation nanoscale devices, but deep insights are required to provide a wealth of research possibilities for the disciplines of chemical, physical, and materials science.

Thus, Spintronics based on 2D materials has a promising and fascinating future!

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors extend their appreciation to the Deanship of Scientific Research at King Khalid University for funding this work through large group Research Project under grant number RGP.2/573/44. This work also supported by the basic research program by ICT of NRF Korea (Grant no: 2022R1F1A1075229).

References

- [1] D. Geng, H.Y. Yang, *Adv. Mater.* 30 (45) (2018) 1800865.
- [2] E. Elahi et al., *Mater. Today Phys.* (2022) 100943.
- [3] M. Iqbal et al., *Superlattice. Microst.* 137 (2020) 106350.
- [4] E. Elahi et al., *Dalton Trans.* 49 (29) (2020) 10017.
- [5] M.F. Khan et al., *ACS Appl. Electron. Mater.* 5 (9) (2023) 5111.
- [6] E. Elahi et al., *J. Mater. Chem. C* (2023).
- [7] E. Elahi et al., *Appl. Mater. Today* 32 (2023) 101846.
- [8] S. Aftab et al., *J. Colloid Interface Sci.* 603 (2021) 511.
- [9] M. Gibertini et al., *Nat. Nanotechnol.* 14 (5) (2019) 408.
- [10] G. Zhang et al., *Compos. Struct.* 318 (2023) 117082.
- [11] R. Ahammed, A. De Sarkar, *Phys. Rev. B* 105 (4) (2022) 045426.
- [12] D. Joksas, et al., *arXiv preprint arXiv:2203.06147* (2022)
- [13] W. Huh et al., *Adv. Mater.* 32 (51) (2020) 2002092.
- [14] Y. Xiao et al., *Chem. Soc. Rev.* (2023).
- [15] J. Khan et al., *SmartMat* 4 (2) (2023) e1156.
- [16] P.V. Pham et al., *Chem. Rev.* 122 (6) (2022) 6514.
- [17] P. Ares, K.S. Novoselov, *Nano Mater. Sci.* 4 (1) (2022) 3.
- [18] Z.L. Lei, B. Guo, *Adv. Sci.* 9 (4) (2022) 2102924.
- [19] E. Elahi et al., *Comput. Mater. Sci.* 213 (2022) 111670.
- [20] J.-U. Lee et al., *Nano Lett.* 16 (12) (2016) 7433.
- [21] M. Bonilla et al., *Nat. Nanotechnol.* 13 (4) (2018) 289.
- [22] D.J. O'Hara et al., *Nano Lett.* 18 (5) (2018) 3125.
- [23] Z. Wang et al., *Nat. Nanotechnol.* 13 (7) (2018) 554.
- [24] D. Ghazaryan et al., *Nat. Electron.* 1 (6) (2018) 344.
- [25] B. Huang et al., *Nature* 546 (7657) (2017) 270.
- [26] A. Barman et al., *J. Phys. Condens. Matter* 33 (41) (2021) 413001.
- [27] Q.H. Wang et al., *ACS Nano* 16 (5) (2022) 6960.
- [28] X. Lin et al., *Nat. Electron.* 2 (7) (2019) 274.
- [29] Y. Zhang et al., in: *2014 Design, Automation & Test in Europe Conference & Exhibition (DATE), IEEE, 2014*, p. 1.
- [30] X. Liu, M.C. Hersam, *Nat. Rev. Mater.* 4 (10) (2019) 669.
- [31] K. Choudhary et al., *npj Comput. Mater.* 6 (1) (2020) 49.
- [32] Z. Huang et al., *Adv. Mater.* 34 (16) (2022) 2110464.
- [33] A. Kimel et al., *J. Phys. D Appl. Phys.* 55 (46) (2022) 463003.
- [34] H. Yang et al., *Micromachines* 13 (1) (2022) 92.
- [35] C.-Z. Chang et al., *Rev. Mod. Phys.* 95 (1) (2023) 011002.
- [36] J. Pan et al., *npj Comput. Mater.* 6 (1) (2020) 152.
- [37] S. Zhang et al., *Nanoscale* 13 (3) (2021) 1398.
- [38] D. Ghader, *Sci. Rep.* 10 (1) (2020) 16733.
- [39] A.V. Chumak et al., *IEEE Trans. Magn.* 58 (6) (2022) 1.
- [40] K.L. Seyler et al., *Nat. Phys.* 14 (3) (2018) 277.
- [41] L. Zhang et al., *Vacuum* (2023) 112299.
- [42] K.S. Burch et al., *Nature* 563 (7729) (2018) 47.
- [43] C. Gong, X. Zhang, *Science* 363 (6428) (2019) eaav4450.
- [44] C. Chang et al., *Acta Phys.-Chim. Sin.* 37 (12) (2021) 2108017.
- [45] K.F. Mak et al., *Nat. Rev. Phys.* 1 (11) (2019) 646.
- [46] J. Yao et al., *Adv. Mater.* (2022) 2200236.
- [47] X. Jiang et al., *Appl. Phys. Rev.* 8 (3) (2021) 031305.
- [48] H. Li et al., *Adv. Mater.* 31 (27) (2019) 1900065.
- [49] R. Ashraf et al., *Results Phys.* 52 (2023) 106755.
- [50] C. Gong et al., *Nature* 546 (7657) (2017) 265.
- [51] Z. Zhang et al., *Nano Lett.* 19 (5) (2019) 3138.
- [52] A.F. May et al., *ACS Nano* 13 (4) (2019) 4436.
- [53] Y. Deng et al., *Nature* 563 (7729) (2018) 94.
- [54] B. Xu et al., *Appl. Phys. Lett.* 116 (5) (2020) 052403.
- [55] M.M. Otrokov et al., *Nature* 576 (7787) (2019) 416.
- [56] Y. Peng et al., *Adv. Mater.* 32 (28) (2020) 2001200.

- [57] K. Kim et al., *Nat. Commun.* 10 (1) (2019) 1.
- [58] S. Kim et al., *2D Mater* 6 (4) (2019).
- [59] M.A. McGuire, *Crystals* 7 (5) (2017) 121.
- [60] S. Jiang et al., *Nat. Nanotechnol.* 13 (7) (2018) 549.
- [61] B. Huang et al., *Nat. Nanotechnol.* 13 (7) (2018) 544.
- [62] S. Jiang et al., *Nat. Mater.* 17 (5) (2018) 406.
- [63] S. Kazim et al., *Nanotechnology* 31 (39) (2020) 395706.
- [64] J. Su et al., *Adv. Sci.* 7 (22) (2020) 2001722.
- [65] V. Nicolosi et al., *Science* 340 (6139) (2013) 1226419.
- [66] L. León-Alcaide et al., *Chem. Commun.* 56 (55) (2020) 7657.
- [67] G.K. Rajendran Nair et al., *Phys. Status Solidi (RRL)–Rapid Res. Lett.* 16 (4) (2022) 2100495.
- [68] H. Jiang et al., *Nano Res.* 14 (2021) 1789.
- [69] Y. Wen et al., *Nano Lett.* 20 (5) (2020) 3130.
- [70] S. Liu et al., *npj 2D Mater. Appl.* 1 (1) (2017) 30.
- [71] P. Li et al., *Sci. Bull.* 65 (13) (2020) 1064.
- [72] C.-S. Lian et al., *Phys. Rev. B* 100 (23) (2019) 235420.
- [73] S. Le et al., *Bull. Am. Phys. Soc.* (2023).
- [74] P. Redhu et al., *Mater. Today: Proc.* (2023).
- [75] J. Hu et al., *J. Alloy. Compd.* 911 (2022) 164830.
- [76] B. Chen et al., *Nat. Commun.* 10 (1) (2019) 4469.
- [77] I. Žutić et al., *Mater. Today* 22 (2019) 85.
- [78] A.I. Buzdin, *Rev. Mod. Phys.* 77 (3) (2005) 935.
- [79] J. Stahn et al., *Phys. Rev. B* 71 (14) (2005) 140509.
- [80] I. Vobornik et al., *Nano Lett.* 11 (10) (2011) 4079.
- [81] V. Korenev et al., *Nat. Commun.* 3 (1) (2012) 1.
- [82] B. Scharf et al., *Phys. Rev. Lett.* 119 (12) (2017) 127403.
- [83] M. Li et al., *J. Mater. Res. Technol.* (2022).
- [84] C. Lee et al., *Nat. Commun.* 7 (1) (2016) 1.
- [85] S. Eremeev et al., *Phys. Rev. B* 88 (14) (2013) 144430.
- [86] Y. Gao et al., *IEEE J. Emerg. Select. Top. Power Electron.* 10 (2) (2021) 1671.
- [87] S. Datta, B. Das, *Appl. Phys. Lett.* 56 (7) (1990) 665.
- [88] I. Žutić et al., *Rev. Mod. Phys.* 76 (2) (2004) 323.
- [89] D.D. Awschalom, M.E. Flatté, *Nat. Phys.* 3 (3) (2007) 153.
- [90] S. Sugahara, J. Nitta, *Proc. IEEE* 98 (12) (2010) 2124.
- [91] W. Yan et al., *Nat. Commun.* 7 (1) (2016) 1.
- [92] A. Dankert, S.P. Dash, *Nat. Commun.* 8 (1) (2017) 1.
- [93] H.C. Koo et al., *Science* 325 (5947) (2009) 1515.
- [94] P. Chuang et al., *Nat. Nanotechnol.* 10 (1) (2015) 35.
- [95] Y. Lu et al., *Smart Mater. Struct.* 32 (1) (2022) 017001.
- [96] F. Hellman et al., *Rev. Mod. Phys.* 89 (2) (2017) 025006.
- [97] C.-C. Chiang et al., *Appl. Phys. Rev.* 8 (2021) 2.
- [98] H. Yang et al., *Nature* 606 (7915) (2022) 663.
- [99] H. Lin et al., *Micromachines* 13 (2) (2022) 319.
- [100] V. Krizakova et al., *J. Magn. Magn. Mater.* 562 (2022) 169692.
- [101] J. Shan, *Bull. Am. Phys. Soc.* (2020) 65.
- [102] L. Zhang et al., *Appl. Phys. Rev.* 8 (2) (2021) 021308.
- [103] M. Och et al., *Nanoscale* 13 (4) (2021) 2157.
- [104] K. Momma, F. Izumi, in: *atelier de diffraction sur poudre Seventh Canadian Powder Diffraction Workshop*, 2006, p 106
- [105] K. Momma, F.J. Izumi, *J. o. A. c.* 41 (3) (2008) 653.
- [106] Y. He et al., *IEEE Trans. Antennas Propag.* (2023).
- [107] P. Nieves et al., *Comput. Mater. Sci* 168 (2019) 188.
- [108] A. Jain et al., *APL Mater.* 1 (2013) 1.
- [109] A. Vaitkus, et al., in: *Acta Crystallographica A-Foundation and Advances, Int Union crystallography 2 Abbey Sq, Chester, CH1 2HU, England, 2021, vol. 77, pp C814*
- [110] S. Gražulis et al., *Handbook of Materials Modeling: Methods: Theory and Modeling*, 2020, 1863
- [111] S. Gražulis et al., *Mater. Informat.: Methods Tools Appl.* (2019) 1.
- [112] A. Vaitkus et al., *J. Appl. Cryst.* 54 (2) (2021) 661.
- [113] R. Cheng et al., *Adv. Mater.* 29 (35) (2017) 1703122.
- [114] A. Gupta et al., *Prog. Mater. Sci.* 73 (2015) 44.
- [115] M. Iqbal et al., *Superlattice. Microst.* 135 (2019) 106247.
- [116] M.W. Iqbal et al., *Microelectron. Eng.* 265 (2022) 111885.
- [117] M.W. Iqbal et al., *J. Mater. Sci. Mater. Electron.* 34 (7) (2023) 677.
- [118] A.K. Geim, K.S. Novoselov, *Nat. Mater.* 6 (3) (2007) 183.
- [119] A. Kumar et al., *J. Magn. Magn. Mater.* 542 (2022) 168507.
- [120] K. Novoselov, A.C.J.P. Neto, S. 2012 (T146) (2012) 014006.
- [121] R.A. Patil et al., *Mater. Today Phys.* 12 (2020) 100174.
- [122] Y. Huang et al., *Nat. Commun.* 11 (1) (2020) 2453.
- [123] C.-T. Kuo et al., *Sci. Rep.* 6 (1) (2016) 20904.
- [124] G. Wang et al., *ACS Nano* 10 (2) (2016) 1788.
- [125] M. Huang et al., *Nat. Commun.* 13 (1) (2022) 5369.
- [126] F.M. Oliveira et al., *FlatChem* 32 (2022) 100334.
- [127] P. Kumar et al., *Phys. Rev. Appl* 18 (6) (2022) L061002.
- [128] T. Zhang et al., *Adv. Sci.* 9 (11) (2022) 2105483.
- [129] S. Jiang et al., *Nat. Electron.* 2 (4) (2019) 159.
- [130] M.W. Iqbal et al., *J. Mater. Sci. Mater. Electron.* 33 (15) (2022) 12416.
- [131] X. Li et al., *Science* 324 (5932) (2009) 1312.
- [132] Q. Ji et al., *Nano Lett.* 13 (8) (2013) 3870.
- [133] T.P. O'Regan et al., *Appl. Phys. Lett.* 111 (2017) 5.
- [134] D. Dumcenco et al., *ACS Nano* 9 (4) (2015) 4611.
- [135] K. Kang et al., *Nature* 520 (7549) (2015) 656.
- [136] Y. Gao et al., *Adv. Mater.* 29 (29) (2017) 1700990.
- [137] H. Yu et al., *ACS Nano* 11 (12) (2017) 12001.
- [138] J. Li et al., *Nature* 579 (7799) (2020) 368.
- [139] L. Meng et al., *Nat. Commun.* 12 (1) (2021) 809.
- [140] B. Li et al., *Nat. Mater.* 20 (6) (2021) 818.
- [141] Q. Ji et al., *Nano Lett.* 17 (8) (2017) 4908.
- [142] M. Hossain et al., *Adv. Mater. Interfaces* 5 (16) (2018) 1800528.
- [143] C. Chen et al., *Adv. Mater.* 34 (2) (2022) 2107512.
- [144] Y. Zhang et al., *Adv. Mater.* 31 (19) (2019) 1900056.
- [145] X. Zhang et al., *Jpn. J. Appl. Phys.* 55 (3) (2016) 033001.
- [146] B. Chen et al., *J. Phys. Soc. Jpn.* 82 (12) (2013) 124711.
- [147] X. Song et al., *J. Mater. Chem. C* 1 (17) (2013) 2952.
- [148] J.-G. Park, *arXiv preprint arXiv:1604.08833*, 2016.
- [149] J.R. Arthur, *Surf. Sci.* 500 (1–3) (2002) 189.
- [150] D.K. Singh, G. Gupta, *Mater. Adv.* 3 (15) (2022) 6142.
- [151] S. Zhang et al., *Mater. Horiz.* 9 (2) (2022) 559.
- [152] W. Ji, *ACS Appl. Electron. Mater.* 4 (2022) 3166.
- [153] J.H. Jeon et al., *ACS Nano* 16 (6) (2022) 8974.
- [154] J. Hauser, *Phys. Rev.* 187 (2) (1969) 580.
- [155] A. Avsar et al., *Nat. Commun.* 5 (1) (2014) 1.
- [156] E. Wolf, *Rep. Prog. Phys.* 41 (9) (1978) 1439.
- [157] M. Kompaniets et al., *J. Supercond. Nov. Magn.* 28 (2) (2015) 431.
- [158] J. Alicea, *Rep. Prog. Phys.* 75 (7) (2012) 076501.
- [159] W. Han, *APL Mater.* 4 (3) (2016) 032401.
- [160] A.K. Geim, I.V. Grigorieva, *Nature* 499 (7459) (2013) 419.
- [161] P. Ajayan et al., *Phys. Today* 69 (9) (2016) 38.
- [162] S. Manzeli et al., *Nat. Rev. Mater.* 2 (8) (2017) 1.
- [163] X. Xu et al., *Nat. Phys.* 10 (5) (2014) 343.
- [164] J. Wang et al., *RSC Adv.* 7 (27) (2017) 16801.
- [165] M.H. Alattas, *Proximity Mechanisms in Graphene: Insights from Density Functional Theory*, 2018.
- [166] B. Hunt et al., *Science* 340 (6139) (2013) 1427.
- [167] W.S. Torres et al., *2D Mater.* 4 (4) (2017) 041008.
- [168] P. Lazić et al., *Phys. Rev. B* 90 (8) (2014) 085429.
- [169] D.K. Efetov et al., *Nat. Phys.* 12 (4) (2016) 328.
- [170] Y. Cao et al., *Nature* 556 (7699) (2018) 43.
- [171] A. Hrabec et al., *Nat. Commun.* 8 (1) (2017) 1.
- [172] P.-J. Hsu et al., *Nat. Nanotechnol.* 12 (2) (2017) 123.
- [173] Y. Wu et al., *Nat. Commun.* 11 (1) (2020) 3860.
- [174] D. Aasen et al., *Phys. Rev. X* 6 (3) (2016) 031016.
- [175] S.D. Sarma, *npj Quantum, et al., Information* 1 (1) (2015) 1.
- [176] M.Z. Hasan, C.L. Kane, *Rev. Mod. Phys.* 82 (4) (2010) 3045.
- [177] S. Bhattacharyya et al., *Adv. Mater.* 33 (33) (2021) 2007795.
- [178] C. Liu et al., *Nat. Mater.* 19 (5) (2020) 522.
- [179] Y. Deng et al., *Science* 367 (6480) (2020) 895.
- [180] R. Watanabe et al., *Appl. Phys. Lett.* 115 (10) (2019) 102403.
- [181] P. Li et al., *Sci. Adv.* 5 (8) (2019) eaaw3415.
- [182] C.O. Avci et al., *Nat. Mater.* 16 (3) (2017) 309.
- [183] Q. Shao et al., *Nat. Commun.* 9 (1) (2018) 1.
- [184] Y. Fan et al., *Nat. Mater.* 13 (7) (2014) 699.
- [185] K. Yasuda et al., *Phys. Rev. Lett.* 119 (13) (2017) 137204.
- [186] M. Mogi et al., *Nat. Commun.* 12 (1) (2021) 1.
- [187] T. Liang et al., *Nat. Phys.* 14 (5) (2018) 451.
- [188] R. Welsch et al., *Bull. Am. Phys. Soc.* (2023).
- [189] S.N. Guin et al., *NPG Asia Mater.* 11 (1) (2019) 16.
- [190] M. Mizuguchi, *Oyo Buturi (Online)* 90 (2) (2021) 78.
- [191] H. Akinaga, *Jpn. J. Appl. Phys.* 59 (11) (2020) 110201.
- [192] J. Xu et al., *Nano Lett.* 19 (11) (2019) 8250.
- [193] M. Mogi et al., *Nat. Commun.* 12 (1) (2021) 1404.
- [194] M. Mogi et al., *APL Mater.* 6 (9) (2018) 091104.
- [195] M. Mogi et al., *Phys. Rev. Lett.* 123 (1) (2019) 016804.
- [196] L. Liu et al., *Phys. Rev. Lett.* 109 (9) (2012) 096602.

- [197] O. Lee et al., *Phys. Rev. B* 89 (2) (2014) 024418.
- [198] I.M. Miron et al., *Nature* 476 (7359) (2011) 189.
- [199] K. Zollner et al., *Phys. Rev. B* 100 (8) (2019) 085128.
- [200] C. Zhao et al., *Nat. Nanotechnol.* 12 (8) (2017) 757.
- [201] D. Zhong et al., *Nat. Nanotechnol.* 15 (3) (2020) 187.
- [202] D. Xiao et al., *Phys. Rev. Lett.* 97 (2) (2006) 026603.
- [203] Y. Pu et al., *Phys. Rev. Lett.* 101 (11) (2008) 117208.
- [204] N. Nagaosa et al., *Rev. Mod. Phys.* 82 (2) (2010) 1539.
- [205] Z. Wang et al., *Phys. Rev. Lett.* 114 (1) (2015) 016603.
- [206] P. Wei et al., *Nat. Mater.* 15 (7) (2016) 711.
- [207] C. Tang et al., *Sci. Adv.* 3 (6) (2017) e1700307.
- [208] W. Zhao et al., *Nat. Mater.* 19 (5) (2020) 503.
- [209] J. Li et al., *Nat. Commun.* 13 (1) (2022) 1.
- [210] M. Lohmann et al., *Nano Lett.* 19 (4) (2019) 2397.
- [211] S. Deng et al., *Appl. Phys. Lett.* 117 (26) (2020) 262402.
- [212] T. Seki et al., *Phys. Rev. B* 103 (2) (2021) L020402.
- [213] P. Sheng et al., *Appl. Phys. Lett.* 116 (14) (2020) 142403.
- [214] J. Wang et al., *Adv. Electron. Mater.* 8 (9) (2022) 2101380.
- [215] W. Zhou et al., *Nat. Mater.* 20 (4) (2021) 463.
- [216] V. Fatemi et al., *Phys. Rev. B* 95 (4) (2017) 041410.
- [217] M.N. Ali et al., *Nature* 514 (7521) (2014) 205.
- [218] Y. Wang et al., *Nat. Commun.* 7 (1) (2016) 1.
- [219] P. Olbrich et al., *Phys. Rev. Lett.* 113 (9) (2014) 096601.
- [220] Y. Zhang et al., *2D Mater.* 5 (4) (2018) 044001.
- [221] L.-K. Shi, J.C. Song, *Phys. Rev. B* 99 (3) (2019) 035403.
- [222] J.-S. You et al., *Phys. Rev. B* 98 (12) (2018) 121109.
- [223] D. MacNeill et al., *Nat. Phys.* 13 (3) (2017) 300.
- [224] D. MacNeill et al., *Phys. Rev. B* 96 (5) (2017) 054450.
- [225] C.M. Acosta et al., *ACS Appl. Mater. Interfaces* 14 (7) (2022) 9418.
- [226] D. Ge et al., *J. Magn. Magn. Mater.* 545 (2022) 168753.
- [227] E. Marchiori et al., *Nat. Rev. Phys.* 4 (1) (2022) 49.
- [228] D. Kumar et al., *Phys. Rep.* 958 (2022) 1.
- [229] P. Cai et al., *Phys. Rev. Lett.* 115 (5) (2015) 057202.
- [230] Y. Zhao et al., *Phys. Rev. B* 92 (4) (2015) 041104.
- [231] I. Pletikosić et al., *Phys. Rev. Lett.* 113 (21) (2014) 216601.
- [232] L. Wang et al., *Nat. Commun.* 6 (1) (2015) 8892.
- [233] K. Kang et al., *Nat. Mater.* 18 (4) (2019) 324.
- [234] J. Fabian et al., *Phys. Rev. B* 66 (16) (2002) 165301.
- [235] S.-J. Gong et al., *Proc. Natl. Acad. Sci.* 115 (34) (2018) 8511.
- [236] P.M. Coelho et al., *J. Phys. Chem. C* 123 (22) (2019) 14089.
- [237] A.O. Fumega et al., *J. Phys. Chem. C* 123 (45) (2019) 27802.
- [238] T. Song et al., *Science* 360 (6394) (2018) 1214.
- [239] D.R. Klein et al., *Science* 360 (6394) (2018) 1218.
- [240] H.H. Kim et al., *Nano Lett.* 18 (8) (2018) 4885.
- [241] L. Britnell et al., *Science* 335 (6071) (2012) 947.
- [242] L. Britnell et al., *Nat. Commun.* 4 (1) (2013) 1794.
- [243] Z. Wang et al., *Nat. Commun.* 9 (1) (2018) 2516.
- [244] F. Matsukura et al., *Nat. Nanotechnol.* 10 (3) (2015) 209.
- [245] E. Elahi et al., *J. Phys. D Appl. Phys.* (2022).
- [246] L. Zhou et al., *InfoMat* (2022) e12371.
- [247] T. Scheike et al., *Appl. Phys. Lett.* 118 (4) (2021) 042411.
- [248] J. Zhou et al., *ACS Appl. Mater. Interfaces* 11 (19) (2019) 17647.
- [249] D. Li, *J. Phys. Chem. C* 125 (29) (2021) 16228.
- [250] H. Lin et al., *ACS Appl. Mater. Interfaces* 12 (39) (2020) 43921.
- [251] E. Elahi et al., *Phys. Status Solidi (RRL)–Rapid Res. Lett.* 17 (2) (2023) 2200212.
- [252] S. Sanvito, *Chem. Soc. Rev.* 40 (6) (2011) 3336.
- [253] I. Bergenti et al., *ACS Appl. Electronic Mater.* 4 (9) (2022) 4273.
- [254] J.M. De Teresa et al., *Science* 286 (5439) (1999) 507.
- [255] F. Djeghloul et al., *Sci. Rep.* 3 (1) (2013) 1.
- [256] M.F. Khan et al., *Nanoscale Res. Lett.* 15 (2020) 1.
- [257] M.F. Khan et al., *Nanoscale* 10 (35) (2018) 16703.
- [258] K. Nagasaka et al., *J. Appl. Phys.* 89 (11) (2001) 6943.
- [259] H.-C. Wu et al., *Sci. Rep.* 3 (1) (2013) 1.
- [260] Z.-M. Liao et al., *Appl. Phys. Lett.* 98 (2011) 5.
- [261] M. Julliere, *Phys. Lett. A* 54 (3) (1975) 225.
- [262] E.Y. Tsybal et al., *J. Phys. Condens. Matter* 15 (4) (2003) R109.
- [263] T.M. Mohiuddin et al., *IEEE Trans. Magn.* 44 (11) (2008) 2624.
- [264] V. Karpan et al., *Phys. Rev. Lett.* 99 (17) (2007) 176602.
- [265] V. Karpan et al., *Phys. Rev. B* 78 (19) (2008) 195419.
- [266] H. Wen et al., *Phys. Rev. Appl.* 5 (4) (2016) 044003.
- [267] V. Zlatko et al., *ACS Nano* 16 (9) (2022) 14007.
- [268] W. Li et al., *Phys. Rev. B* 89 (18) (2014) 184418.
- [269] K. Dolui et al., *Phys. Rev. B* 90 (4) (2014) 041401.
- [270] M. Piquemal-Banci, I., et al., *ACS Nano* 12 (5) (2018) 4712.
- [271] V. Zlatko et al., *ACS Nano* 13 (12) (2019) 14468.
- [272] K. Zhao et al., *J. Magn. Magn. Mater.* 432 (2017) 10.
- [273] L. Xu et al., *Adv. Condens. Matter Phys.* (2017) 2017.
- [274] M.F. Khan et al., *APL Mater.* 8 (2020) 7.
- [275] H.J. Deiseroth et al., *Fe₃GeTe₂ and Ni₃GeTe₂—Two New Layered Transition-Metal Compounds: Crystal Structures, HRTEM Investigations, and Magnetic and Electrical Properties*, Wiley Online Library, 2006.
- [276] K. Kim et al., *Nat. Mater.* 17 (9) (2018) 794.
- [277] S.Y. Park et al., *Nano Lett.* 20 (1) (2019) 95.
- [278] Z. Wang et al., *Nano Lett.* 18 (7) (2018) 4303.
- [279] M. Bibes et al., *Appl. Phys. Lett.* 82 (19) (2003) 3269.
- [280] T. Banerjee et al., *Phys. Rev. B* 81 (21) (2010) 214409.
- [281] E. Cobas et al., *Nano Lett.* 12 (6) (2012) 3000.
- [282] J.-J. Chen et al., *Nat Commun* 4 (1) (2013) 1.
- [283] J.-H. Park, H.-J. Lee, *Phys. Rev. B* 89 (16) (2014) 165417.
- [284] B. Dlubak et al., *ACS Nano* 6 (12) (2012) 10930.
- [285] A.K. Singh, J. Eom, *ACS Appl. Mater. Interfaces* 6 (4) (2014) 2493.
- [286] M.-B. Martin et al., *ACS Nano* 8 (8) (2014) 7890.
- [287] A.L. Friedman et al., *Nat Commun* 5 (1) (2014) 1.
- [288] W. Wang et al., *Nano Lett.* 15 (8) (2015) 5261.
- [289] L. Britnell et al., *Nano Lett.* 12 (3) (2012) 1707.
- [290] T. Roy et al., *Appl. Phys. Lett.* 104 (12) (2014) 123506.
- [291] L. Zhang et al., *J. Phys. Chem. C* 124 (50) (2020) 27429.
- [292] C. Tan et al., *Nat. Commun.* 9 (1) (2018) 1554.
- [293] Z.-S. Zhang, D. Rabson, *J. Appl. Phys.* 95 (2) (2004) 557.
- [294] R. Victora, X. Chen, *IEEE Trans. Magn.* 46 (3) (2010) 702.
- [295] X. Chen, R. Victora, *Appl. Phys. Lett.* 91 (2007) 21.
- [296] H. Dery et al., *IEEE Trans. Electron Devices* 59 (1) (2011) 259.
- [297] W. Zhu et al., *Chin. Phys. Lett.* 39 (12) (2022) 128501.
- [298] H. Lin et al., *Nanoscale* 14 (6) (2022) 2352.
- [299] K.-H. Min et al., *Bull. Am. Phys. Soc.* (2023).
- [300] W. Zhu, et al., *arXiv preprint arXiv:2303.11596*, 2023.
- [301] W. Zhu et al., *Adv. Mater.* 33 (51) (2021) 2104658.
- [302] H.-C. Wu et al., *Sci. Rep.* 5 (1) (2015) 1.
- [303] P. Asshoff et al., *2D Mater.* 4 (3) (2017) 031004.
- [304] H. Zhou et al., *ACS Appl. Mater. Interfaces* 13 (1) (2020) 1214.
- [305] C. Hu et al., *Chin. Phys. B* (2021).
- [306] L. Zhou et al., *InfoMat* 5 (3) (2023) e12371.
- [307] Y. Zhang et al., *Phys. Status Solidi (RRL)–Rapid Res. Lett.* (2023).
- [308] Z. Chen et al., *Nanoscale* 15 (18) (2023) 8447.
- [309] P. Gambardella, I.M. Miron, *Phil. Trans. R. Soc. A: Math. Phys. Eng. Sci.* 369 (1948) (2011) 3175.
- [310] A. Manchon et al., *Rev. Mod. Phys.* 91 (3) (2019) 035004.
- [311] A. Johansen et al., *Phys. Rev. Lett.* 122 (21) (2019) 217203.
- [312] M. Jiang et al., *Nat. Commun.* 10 (1) (2019) 1.
- [313] M. Jiang et al., *Jpn Soc. Appl. Phys.* (2018) 2096.
- [314] C.Y. Wang et al., *Adv. Electron. Mater.* 6 (2) (2020) 1901107.
- [315] L. Tong et al., *Science* 373 (6561) (2021) 1353.
- [316] M. Shashidhara, A. Acharya, in: *Emerging Low-Power Semiconductor Devices*, CRC Press, 2022, p. 273.
- [317] H. Yu et al., *PCCP* 25 (15) (2023) 10991.
- [318] R. Li et al., *Nanoscale* 12 (44) (2020) 22808.
- [319] L. Shen et al., *Appl. Phys. Lett.* 114 (2019) 4.
- [320] L. Liu et al., *Phys. Rev. Lett.* 109 (18) (2012) 186602.
- [321] K. Ando et al., *Phys. Rev. Lett.* 101 (3) (2008) 036601.
- [322] A. Manchon, S. Zhang, *Phys. Rev. B* 78 (21) (2008) 212405.
- [323] V.M. Edelstein, *Solid State Commun.* 73 (3) (1990) 233.
- [324] Q. Shao et al., *Nano Lett.* 16 (12) (2016) 7514.
- [325] V. Ostwal et al., *Adv. Mater.* 32 (7) (2020) 1906021.
- [326] C. Safeer et al., *Nano Lett.* 19 (12) (2019) 8758.
- [327] B. Zhao et al., *Physical Review Research* 2 (1) (2020) 013286.
- [328] J. Hidding, M.H. Guimarães, *Front. Mater.* 7 (2020) 594771.
- [329] S. Husain et al., *Appl. Phys. Rev.* 7 (4) (2020) 041312.
- [330] A.M. Hoque et al., *Phys. Rev. Res.* 2 (3) (2020) 033204.
- [331] N.H.D. Khang, et al. (2018) 17(9) 808
- [332] M. De et al., *Nat. Mater.* 17 (9) (2018) 800.
- [333] B. Zhao, et al. (2020) 32(38) 2000818
- [334] P. Li et al., *Nat. Commun.* 9 (1) (2018) 3990.
- [335] B. Zhao et al., *Adv. Mater.* 32 (38) (2020) 2000818.
- [336] S.-H.-C. Baek et al., *Nat. Mater.* 17 (6) (2018) 509.
- [337] V. Amin, M. Stiles, *Phys. Rev. B* 94 (10) (2016) 104420.
- [338] X. Wang et al., *Sci. Adv.* 5 (8) (2019) eaaw8904.

- [339] M. Alghamdi et al., *Nano Lett.* 19 (7) (2019) 4400.
- [340] V. Gupta et al., *Nano Lett.* 20 (10) (2020) 7482.
- [341] N. Ito et al., *Phys. Rev. B* 100 (6) (2019) 060402.
- [342] T. Liu et al., *Phys. Rev. B* 101 (20) (2020) 205407.
- [343] K. Garello, et al., in: *APS March Meeting Abstracts*, 2012, vol. 2012, p. P15.010
- [344] S. Emori et al., *Nat. Mater.* 12 (7) (2013) 611.
- [345] G. Beach, *Concepts Spintron.* (2013) 9.
- [346] C. Zhang et al., *Appl. Phys. Lett.* 107 (1) (2015) 012401.
- [347] C.O. Avci et al., *Phys. Rev. B* 90 (22) (2014) 224427.
- [348] M. Hayashi et al., *Phys. Rev. B* 89 (14) (2014) 144425.
- [349] N. Roschewsky et al., *Phys. Rev. B* 99 (19) (2019) 195103.
- [350] N. Roschewsky, et al., in: *APS March Meeting Abstracts*, 2019, vol. 2019, p P41.008
- [351] L. Zhu et al., *Phys. Rev. Appl* 15 (2) (2021) 024059.
- [352] W. Li, et al., *arXiv preprint arXiv:2304.10718*, 2023.
- [353] I. Shin et al., *Adv. Mater.* 34 (8) (2022) 2101730.
- [354] J.M. Daughton, J.S. Huang, *Google Patents* (1988).
- [355] S. Bertolazzi et al., *Adv. Mater.* 31 (10) (2019) 1806663.
- [356] N. Tanaka, *MRS Bull.* (2018) 43.
- [357] A. Kostrov, V. Stempitsky, in: *The 2012 International Conference on Advanced Technologies for Communications*, IEEE, 2012, p. 9.
- [358] S. Alam et al., *Sci. Rep.* 11 (1) (2021) 1.
- [359] B. Dieny et al., *Nat. Electron.* 3 (8) (2020) 446.
- [360] S. Ikegawa et al., *IEEE Trans. Electron Devices* 67 (4) (2020) 1407.
- [361] A.C. Ferrari et al., *Nanoscale* 7 (11) (2015) 4598.
- [362] S. Roche et al., *2D Mater.* 2 (3) (2015) 030202.
- [363] J.F. Sierra et al., *Nat. Nanotechnol.* 16 (8) (2021) 856.
- [364] I. Asselberghs et al., *2020 IEEE International Electron Devices Meeting (IEDM)*, IEEE, 2020. pp 40.2. 1.
- [365] D. Akinwande et al., *Nature* 573 (7775) (2019) 507.
- [366] D.C. Ralph, M.D. Stiles, *J. Magn. Magn. Mater.* 320 (7) (2008) 1190.
- [367] L. Liu et al., *Science* 336 (6081) (2012) 555.
- [368] D. Pesin, A.H. MacDonald, *Nat. Mater.* 11 (5) (2012) 409.
- [369] S. Fukami et al., *Nat. Nanotechnol.* 11 (7) (2016) 621.
- [370] Y. Zhou, S.J. Ramanathan, P. o. t. I. 103 (8) (2015) 1289.
- [371] H.H. Kim et al., *Adv. Mater.* 32 (2) (2020) 1905433.
- [372] Z. Yang et al., *Annu. Rev. Mat. Res.* 41 (1) (2011) 337.
- [373] K. Claborn et al., *J. Am. Chem. Soc.* 125 (48) (2003) 14825.
- [374] F. Wu et al., *Adv. Mater.* (2023) 2211653.
- [375] Y. Wu et al., *Nat. Rev. Mater.* 8 (1) (2023) 41.
- [376] Y. Qi et al., *Adv. Mater.* 35 (12) (2023) 2205714.
- [377] Y.-C. Hsieh et al., *Bull. Am. Phys. Soc.* (2023).
- [378] T. Song et al., *Nat. Mater.* 18 (12) (2019) 1298.
- [379] R. Olmos et al., *J. Phys. Chem. C* (2023).
- [380] M. Hong et al., *Dalton Trans.* 52 (21) (2023) 7290.
- [381] J. Hu et al., *Appl. Surf. Sci.* 560 (2021) 149858.
- [382] N. Miao et al., *J. Am. Chem. Soc.* 140 (7) (2018) 2417.
- [383] T. Xiao et al., *Chem. Phys.* 513 (2018) 182.
- [384] A. Nair et al., *PCCP* 22 (22) (2020) 12806.
- [385] Y. Cao et al., *Phys. Rev. Appl* 17 (5) (2022) L051001.
- [386] Z.-A. Wang et al., *Nano Lett.* 23 (2) (2023) 710.
- [387] S. Gražulis et al., *J. Appl. Cryst.* 42 (4) (2009) 726.
- [388] X. Wang et al., *ACS Nano* 15 (10) (2021) 16236.
- [389] S. Yin et al., *Chin. Phys. B* 30 (2) (2021) 027505.
- [390] J. Yi et al., *2D Mater.* 4 (1) (2016) 011005.
- [391] S. Zhang et al., *Phys. Lett. A* 396 (2021) 127219.
- [392] C. Lane, J.-X. Zhu, *Phys. Rev. B* 102 (7) (2020) 075124.
- [393] M. Mi et al., *Adv. Funct. Mater.* 32 (29) (2022) 2112750.
- [394] Y. Takano et al., *J. Magn. Magn. Mater.* 272 (2004) E593.
- [395] M. Bhide et al., *C. R. Acad. Sci.-Ser. IIC-Chem.* 4 (3) (2001) 189.
- [396] Y. Zhang et al., *Small Methods* 5 (4) (2021) 2001068.
- [397] H. Zhang et al., *PCCP* 23 (16) (2021) 9679.
- [398] Q.-Y. Xie et al., *Chin. Phys. B* 28 (5) (2019) 056102.
- [399] N. Ubrig et al., *2D Materials* 7 (1) (2019) 015007.
- [400] D. Baral et al., *PCCP* 23 (5) (2021) 3225.
- [401] H. Wang et al., *Appl. Phys. Lett.* 117 (2020) 8.
- [402] W.-N. Ren et al., *Sci. Rep.* 11 (1) (2021) 1.
- [403] C. Song et al., *Phys. Rev. B* 99 (21) (2019) 214435.
- [404] N. Sivadas et al., *Phys. Rev. B* 91 (23) (2015) 235425.
- [405] A. Panfilov, G. Grechnev, *Low Temp. Phys.* 47 (10) (2021) 863.
- [406] S.M. Wu et al., *Phys. Rev. Lett.* 116 (9) (2016) 097204.
- [407] H. Zhang et al., in: *Exploration*, Wiley Online Library, 2021, p. 20210010.
- [408] J.B. Lee et al., *Phys. Rev. B* 86 (22) (2012) 224407.
- [409] C. Sürgers et al., *Phys. Rev. B* 90 (10) (2014) 104421.
- [410] T. Saito, D. Nishio-Hamane, *Phys. B Condens. Matter* 603 (2021) 412761.
- [411] Y. Sun et al., *AIP Adv.* 11 (3) (2021) 035138.
- [412] D.C. Freitas et al., *J. Phys. Condens. Matter* 27 (17) (2015) 176002.
- [413] X. Zhang et al., *Nat. Commun.* 12 (1) (2021) 1.
- [414] G. Xiao et al., *J. Mater. Chem. C* 10 (46) (2022) 17665.
- [415] H. Xiao et al., *Adv. Mater. Interfaces* 9 (30) (2022) 2201353.
- [416] Y. Tian et al., *2D Materials* 3 (2) (2016) 025035.
- [417] Z. Fei et al., *Nat. Mater.* 17 (9) (2018) 778.
- [418] M. Seol et al., *Adv. Mater.* 32 (42) (2020) 2003542.
- [419] J. Costa et al., *IEEE Trans. Magn.* 51 (11) (2015) 1.
- [420] S. Sato et al., *Jpn. J. Appl. Phys.* 55 (4S) (2016) 04EE05.
- [421] M. Piquemal-Banci et al., *J. Phys. D Appl. Phys.* 50 (20) (2017) 203002.
- [422] K. Kang et al., *Nature* 550 (7675) (2017) 229.
- [423] P.U. Asshoff et al., *Nano Lett.* 18 (11) (2018) 6954.
- [424] M.Z. Iqbal et al., *Sci. Rep.* 12 (6) (2016) 21038.